An Interactive Education Equality system of India

Discipline of CSE

Lovely School of Computer Science &
Engineering Lovely Professional University,
Phagwara

INT232: DATA SCIENCE TOOLBOX — R PROGRAMMING

PROJECT REPORT

(Project Semester August-January 2026)

Education Equality in India

A State Wise-Study Literacy, School Enrolment, Infrastructure and Gender Equality.

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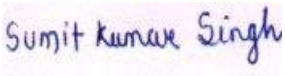
DECLARATION

I hereby declare that the project report entitled “Education Equality in India: A State-wise Study of Literacy, Student Enrolment, Gender Parity and School Infrastructure” has been prepared by me as part of the academic requirements of INT232: Data Science Toolbox — R Programming.

The report documents the data preparation, manipulation, exploratory analysis, descriptive statistical calculations, visualisations and dashboard development carried out for this project using R and its associated packages. The datasets are drawn from the Census of India 2011 literacy tables and UDISE+ school education statistics for 2020–21, 2021–22 and 2023–24. The reference year of each dataset is stated wherever its results are discussed.

I understand that the analyses are descriptive and exploratory. I have not represented association as causation, have not combined incompatible reference years into a single time series, and have identified illustrative or simulated syllabus demonstrations separately from empirical education findings.

All external data sources, software packages and documentation used in the project are acknowledged in the references. The report is submitted for academic evaluation and does not claim to be an official government assessment.

Date: 08/11/2026 Signature:_ 

Registration No.: 62600394

Name: Sumit Kumar SIngh

CERTIFICATE

This is to certify that **Sumit Kumar SIngh** bearing Registration No. 62600394 has completed the INT232 (Data Science Toolbox — R Programming) project titled "Education Equality in India: A State-wise Study of Literacy, Student Enrolment, Gender Parity and School Infrastructure" under my guidance and supervision.

To the best of my knowledge, the present work is the result of the student's development, effort and study. The project applies R programming for importing and cleaning structured datasets, data manipulation, exploratory data analysis, descriptive statistical analysis, graphical visualisation and interactive dashboard presentation.

The report includes the analysis of literacy indicators from Census 2011, enrolment indicators from UDISE+ 2020–21 and 2021–22, and school infrastructure indicators from UDISE+ 2023–24. The interpretations are limited to the definitions, reference periods and coverage of these source datasets.

Signature and Name of the Supervisor: _____

Dr. Tanima Thakur

Designation of the Supervisor: Associate Professor

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Date: 08/11/2026

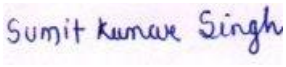
ACKNOWLEDGEMENT

I express my sincere gratitude to Lovely Professional University and the faculty members of the School of Computer Science and Engineering for providing the academic environment and learning resources required to complete this R Programming minor project.

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Finally, I thank my peers and family for their support during the completion of this project.



Student Signature

EXECUTIVE SUMMARY

Education equality is multidimensional. It involves not only whether children are enrolled in school, but also whether girls and boys participate equitably, whether literacy outcomes are balanced, and whether schools have sufficient teachers and basic institutional capacity. This project uses R to organise and examine selected indicators across Indian States and Union Territories.

The study uses four principal source files: Census 2011 literacy data; UDISE+ enrolment tables for 2020–21 and 2021–22; and UDISE+ school infrastructure statistics for 2023–24. The cleaned state-level analysis contains approximately 31 literacy records and 36 enrolment/infrastructure records after excluding the national aggregate row. The datasets are not a unified longitudinal panel: they differ in reference year, definition and coverage.

Study component	Source / scope	Role in the project
Literacy	Census 2011; male, female and persons literacy	Compare literacy levels and gender gaps
Enrolment	UDISE+ 2020–21 and 2021–22; boys, girls and totals	Compare participation by gender and school stage
Infrastructure	UDISE+ 2023–24; schools, enrolment, teachers, PTR	Examine distribution and selected resource indicators
Dashboard	R Shiny / Plotly visualisations	Support interactive exploration and communication

Based on the supplied raw files, the infrastructure dataset contains 36 State/UT records after removal of the India aggregate. The corresponding totals are 1,471,891 schools, 248,045,828 enrolments and 9,807,600 teachers. The simple, unweighted mean of the State/UT pupil–teacher ratio values is 21.2; this is not a pupil-weighted national PTR.

The project demonstrates data cleaning and preprocessing, data manipulation with tidyverse, exploratory data analysis, descriptive statistical summaries, basic and advanced visualisations, interactive Plotly graphics and dashboard presentation. The report also distinguishes the word cloud's illustrative vocabulary and the Bollinger Band's simulated series from evidence-based education findings.

Figure: Bollinger Band Visualization — 10-period moving average with two standard deviations.

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ABBREVIATIONS AND PROJECT SCOPE

Abbreviation	Meaning
Census	Population Census of India
UDISE+	Unified District Information System for Education Plus
UT	Union Territory
PTR	Pupil–Teacher Ratio
EDA	Exploratory Data Analysis
R	Statistical programming language and environment
RStudio	Integrated development environment for R
CSV	Comma-Separated Values
NA	Missing value in R
UT	Union Territory
ggplot2	Grammar of Graphics plotting package
Plotly	Interactive graphing library
Shiny	R framework for interactive web applications
DT	R interface to the DataTables JavaScript library

Scope: The report is a State/UT-level descriptive study. It does not estimate individual-level learning outcomes, causal effects of school resources, or district-level disparities. Literacy, enrolment and infrastructure indicators are analysed within their own source periods.

1. INTRODUCTION

1.1 Background

Education is a central contributor to social development, economic opportunity and civic participation. In a country with substantial geographic, demographic and socio-economic diversity, national averages can conceal considerable differences between States and Union Territories. A State may have a relatively high literacy rate while still exhibiting a gender gap; another may have a large school network but face challenges in teacher distribution or school functionality.

The idea of education equality therefore extends beyond access alone. Access can be examined through enrolment, foundational opportunity through school availability and teachers, and longer-term educational attainment through literacy. Gender equality is a cross-cutting dimension because differences between male and female indicators can reveal structural barriers that aggregate totals obscure.

This project applies R programming to a set of official education indicators. The goal is to produce a reproducible workflow that transforms raw tabular data into interpretable summaries, visualisations and an interactive dashboard.

1.2 Motivation

The datasets are published in different years and formats, with long variable names and separate tables for literacy, enrolment and school infrastructure. Reading them independently is straightforward, but comparing them responsibly requires consistent state labels, numeric conversion, aggregate-row handling and explicit documentation of their reference periods.

1.3 Problem Statement

The project addresses the problem of making multiple public education datasets understandable and comparable without incorrectly treating them as one uniform dataset. The raw files contain national aggregate rows alongside State/UT observations, long column names and measures that represent different concepts. A direct join or comparison without considering year, unit and denominator can lead to misleading conclusions.

- How do male, female and persons literacy indicators vary across States/UTs in Census 2011?
- How is total school enrolment distributed between boys and girls in UDISE+ 2020–21 and 2021–22?
- How do school counts, enrolment, teacher counts and pupil–teacher ratios vary across States/UTs in UDISE+ 2023–24?
- What patterns become more visible through state-wise charts, heat maps and interactive plots?
- How can the analysis be presented in an accessible dashboard while retaining data-source context?

1.4 Objectives

1. Import and inspect the literacy, enrolment and infrastructure datasets.
2. Clean state labels, check missing values and duplicates, and exclude India aggregate rows from State/UT comparisons.
3. Use R data manipulation functions to derive gender gaps, girls' enrolment shares and state rankings.
4. Conduct exploratory and descriptive statistical analysis.
5. Create basic and advanced charts using ggplot2 and related R packages.
6. Build interactive visualisations using Plotly and present results through a dashboard.
7. Document key findings, limitations and future improvements.

1.5 Scope and INT232 Syllabus Alignment

Course / project requirement	Implementation in this project
Importing data	readr::read_csv() for CSV sources
Data structures	Vectors, data frames and tibbles
Data cleaning	trimws(), as.numeric(), is.na(), distinct(), filter()
Data manipulation	dplyr verbs, mutate(), arrange(), summarise(), group_by()
Functions and formulas	Derived gender gap, girls' share and summary functions
Basic visualisation	Scatter, line, bar and pie charts
Advanced visualisation	Heat map, donut, word cloud and Bollinger Band demonstration
Dynamic / interactive plots	ggplot2 and Plotly / ggplotly()
Dashboard	R Shiny application; separate R Markdown Flexdashboard deliverable where rendered

A syllabus demonstration is not automatically an empirical finding. The word cloud uses illustrative education-related terms where a source corpus is not available. The Bollinger Band uses simulated values to demonstrate a rolling mean and standard-deviation envelope. These are clearly labelled in the analysis chapter.

2. SOURCE OF DATASET

2.1 Dataset Provenance

The study combines public education statistics from two major source families. Literacy indicators are taken from the Census of India 2011 table supplied with the project. School enrolment and infrastructure indicators are taken from UDISE+ tables for the academic years 2020–21, 2021–22 and 2023–24.

File used	Rows × columns in raw file	Reference period	Principal use
literacy_census_2011.csv	32 × 11	Census 2011	Male, female and persons literacy
UDISE_2020_21_Table_3.17.csv	37 × 94	2020–21	Enrolment by gender and school stage
UDISE_2021_22_Table_5.17_2.csv	37 × 94	2021–22	Enrolment by gender and school stage
udise_2023_24.csv	37 × 12	2023–24	School, enrolment, teacher and PTR indicators

The data are State/UT aggregates, not individual student records. Their interpretation is consequently ecological: differences between State-level values do not establish the experience or outcome of every student within that State.

2.2 Dataset Schema and Analytical Grain

Dataset	Key dimensions	Variables used
Literacy	State/UT; Census 2011	Persons - All; Male - All; Female - All
Enrolment 2020–21	State/UT; academic year	Boys, Girls, Total for all ages and school stages
Enrolment 2021–22	State/UT; academic year	Boys, Girls, Total for all ages and school stages
Infrastructure 2023–24	State/UT; academic year	Schools, enrolments, teachers, PTR, zero-enrolment and single-teacher indicators

The State/UT field is the common geographic key used for within-period comparisons. It is not sufficient by itself to justify merging all datasets into a single panel: source years differ, some territorial names or administrative classifications may have changed, and the measures describe different concepts.

2.3 Reference Period Compatibility

The literacy values refer to 2011, enrolment values to 2020–21 or 2021–22, and infrastructure values to 2023–24. A chart that places these indicators together must label both periods. The female literacy versus girls' enrolment-share scatter plot is a cross-period exploratory comparison, not a same-year relationship or evidence that one measure caused the other.

2.4 Data Dictionary and Measurement

Variable	Meaning	Unit / interpretation
Male	Male literacy rate	Percent
Female	Female literacy rate	Percent
Persons / Total	Persons literacy rate	Percent
Gender_Gap_pp	Male literacy minus female literacy	Percentage points
Boys	Boys enrolled	Number of students
Girls	Girls enrolled	Number of students
Girls_Share_pct	Girls / total enrolment × 100	Percent of reported enrolment
Total_Schools	Number of schools	Count
Total_Enrolments	Reported student enrolments	Count
Total_Teachers	Reported teachers	Count
PTR	Pupil–teacher ratio	Pupils per teacher (source definition)
Zero_Enrolment_Schools	Schools reporting zero enrolments	Count
Single_Teacher_Schools	Schools reporting a single teacher	Count

A percentage-point gap is the arithmetic difference between two percentages. For example, a male literacy rate of 80% and female literacy rate of 70% gives a gap of 10 percentage points, not a 10% relative difference.

3. EDA PROCESS

3.1 Modular System Architecture and Analytical Pipeline

The Education Equality in India project is designed around a structured R-based analytical workflow. The workflow transforms raw educational datasets into cleaned analytical tables, performs exploratory data analysis, generates visualizations and presents the results through an interactive Shiny dashboard.

The analytical workflow is divided into the following major components:

1. Data Import and Configuration

The data-import component reads the educational datasets from CSV files using `readr::read_csv()`. It establishes the input paths, loads the required R packages and prepares the datasets for further processing.

Primary datasets:

- `cleaned_literacy_2011.csv`
- `cleaned_2020_21.csv`
- `cleaned_2021_22.csv`
- `cleaned_2023_24.csv`
- `infrastructure_analysis_2023_24.csv`
- `infrastructure_summary.csv`

2. Data Cleaning and Preprocessing

This component performs data validation, missing-value inspection, duplicate checking, state-name standardization, national aggregate removal and variable selection.

It ensures that the data is consistent before conducting statistical analysis.

3. Data Analysis and Feature Engineering

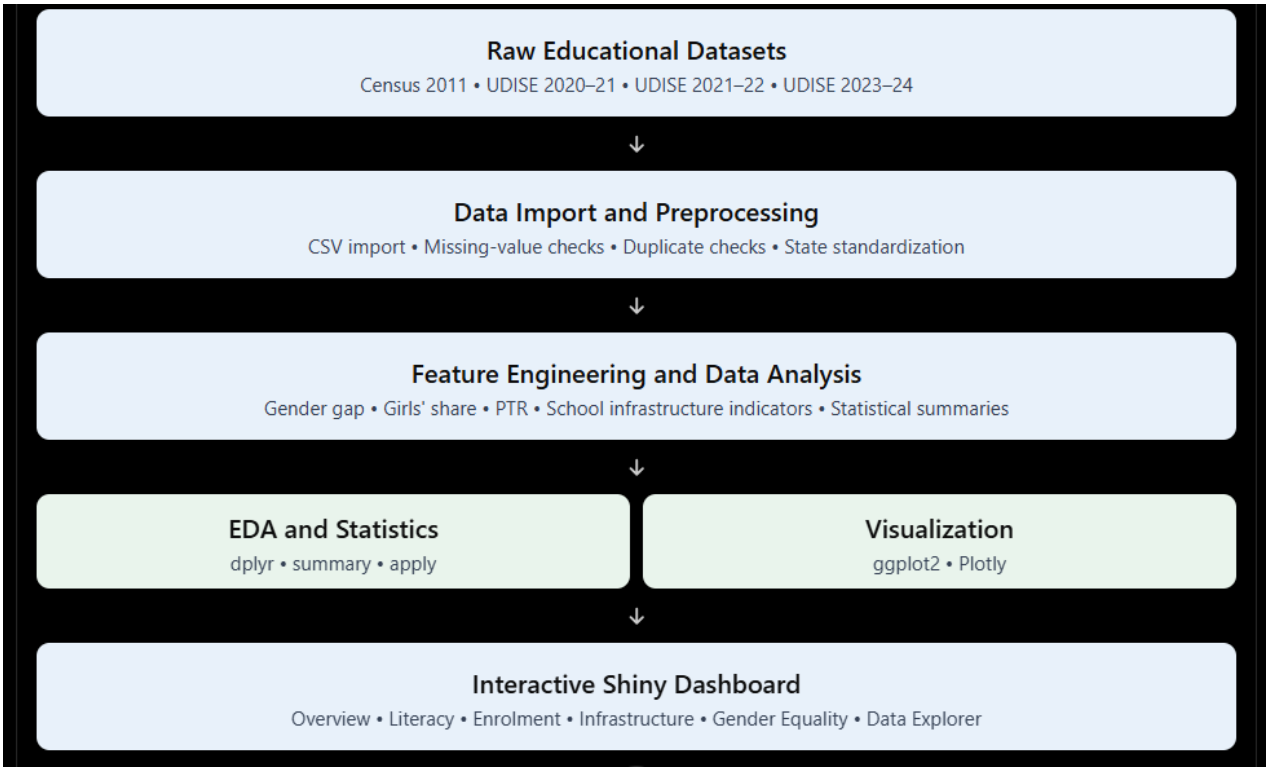
This component calculates additional indicators from the original educational variables. These include gender literacy gap, girls' enrolment share, average enrolment per school, average teachers per school and infrastructure percentages.

It also performs descriptive statistical analysis and state-wise comparisons.

4. Visualization and Dashboard

This component generates static and interactive visualizations using `ggplot2` and `plotly`. The results are integrated into the Shiny dashboard using `shinydashboard`, `DT` and related packages.

Figure 2: Education Equality in India — R Analytical Workflow



3.2 Systematic Data Cleaning and Data Validation

The data-cleaning process is an important stage of the Education Equality in India project. The original datasets contain educational information from different years and sources. Before performing statistical analysis, the datasets are checked and standardized to ensure consistency and reliability.

The preprocessing workflow consists of the following steps.

Step 1: CSV Import and Structural Inspection

The datasets are imported into R using `readr::read_csv()`.

The following functions are used to inspect the imported data:

R function	Purpose
`read_csv()`	Imports CSV files
`dim()`	Returns rows and columns
`head()`	Displays initial observations
`names()`	Displays column names
`str()`	Checks data structure

`glimpse()` Provides a compact data overview

`summary()` Generates descriptive summaries

The dimensions confirmed during the project include:

Dataset	Rows	Columns
cleaned_2020_21.csv	37	94
cleaned_2021_22.csv	37	94
cleaned_2023_24.csv	37	12
cleaned_literacy_2011.csv	32	11

These dimensions include the national aggregate row where present.

Step 2: Missing-Value Detection

Missing values are checked using:

```
colSums(is.na(data))
```

The purpose of this operation is to identify columns containing unavailable observations.

The previously reported cleaning checks found zero missing values in the processed datasets.

Missing values must be handled carefully. A missing educational indicator should not automatically be replaced with zero because zero and unavailable data have different meanings.

Step 3: Duplicate Record Detection

Duplicate observations can affect state-wise summaries and visualizations.

The following functions are used:

```
duplicated(data)
```

and

```
distinct(data)
```

The project cleaning checks did not remove duplicate observations from the processed datasets.

Step 4: State-Name Standardization

State names are standardized to avoid problems caused by leading or trailing spaces.

Example:

```
data$State <- trimws(data$State)
```

This is important because inconsistent state names can create duplicate categories in charts or prevent records from matching during data joins.

Step 5: Removal of National Aggregate

The datasets contain a national aggregate record labelled `INDIA`.

Since the project focuses on comparisons between individual states and union territories, this record is excluded from state-wise analysis.

```
data <- data %>%
  filter(toupper(trimws(State)) != "INDIA")
```

After removing the national aggregate:

- Literacy data: 31 state/UT observations.
- Enrolment data: 36 state/UT observations.
- Infrastructure analysis: 36 state/UT observations.

Step 6: Variable Selection and Renaming

Only the relevant variables are selected for each analytical module.

For literacy analysis:

Original column	Analysis column
State	State
Male - All	Male
Female - All	Female
Persons - All	Total

For enrolment analysis:

Original column	Analysis column
State	State
Enrolment...Boys	Boys
Enrolment...Girls	Girls

Enrolment...Total Total

For infrastructure analysis, the existing descriptive column names are retained.

Step 7: Data Validation and Final Storage

After cleaning, the datasets are checked for:

- Correct column names.
- Appropriate numeric data types.
- Consistent state labels.
- Non-negative school and enrolment counts.
- Correct exclusion of national aggregates.
- Consistency of calculated indicators.

The cleaned datasets are stored in the `data/processed/` directory and reused by the visualization and dashboard components.

3.3 Feature Engineering and Derived Educational Indicators

Feature engineering is the process of creating new analytical variables from existing dataset columns.

In this project, the original literacy, enrolment and infrastructure datasets are transformed into additional indicators to support deeper educational analysis.

The major derived feature groups are:

A. Gender Literacy Gap

The gender literacy gap measures the difference between male and female literacy rates.

Mathematical formulation:

\[Gender\ Gap = Male\ Literacy-Female\ Literacy \]

R implementation:

```
literacy_data <- literacy_data %>%
  mutate(
    Gender_Gap = Male - Female
  )
```

Analytical purpose:

- Identifies differences between male and female literacy.
- Supports state-wise gender equality analysis.
- Helps identify states with relatively large literacy gaps.
- Supports gender-gap bar charts and heat maps.

A positive value indicates that male literacy is higher than female literacy.

The unit is percentage points, not percent change.

B. Girls' Enrolment Share

Girls' enrolment share measures the proportion of total enrolment represented by girls.

Mathematical formulation:

\[Girls\ Share(\%) = \frac{Girls\ Enrolment}{Total\ Enrolment}\times100 \]

R implementation:

```
enrolment_data <- enrolment_data %>%
  mutate(
    Girls_Share = (Girls / Total) * 100
  )
```

Analytical purpose:

- Measures girls' participation in enrolment.
- Supports comparison across states and union territories.
- Provides a descriptive gender-participation indicator.
- Can be compared with a 50% parity reference line.

C. Enrolment Gender Difference

This feature measures the absolute difference between boys' and girls' enrolment.

\[Enrolment\ Gap = Boys-Girls \]

R implementation:

```
enrolment_data <- enrolment_data %>%
  mutate(
    Gender_Gap = Boys - Girls
  )
```

A positive value means boys' enrolment is greater than girls' enrolment.

D. Average Enrolment per School

This indicator measures the average number of students represented per school.

\[Average\ Enrolment = \frac{Total\ Enrolments}{Total\ Schools} \]

R implementation:

```
infrastructure_data <- infrastructure_data %>%
  mutate(
    Avg_Enrolments =
      Total_Enrolments / Total_Schools
  )
```

Analytical purpose:

- Helps examine differences in school size.
- Supports comparisons between states.
- Provides context for the distribution of student enrolment.

E. Average Teachers per School

This indicator calculates the average number of teachers per school.

$$ Average\ Teachers = \frac{Total\ Teachers}{Total\ Schools} $$

R implementation:

```
infrastructure_data <- infrastructure_data %>%
  mutate(

    Avg_Teachers =
      Total_Teachers / Total_Schools
  )
```

F. Pupil-Teacher Ratio

The pupil-teacher ratio represents the number of students per teacher.

$$ PTR = \frac{Total\ Enrolments}{Total\ Teachers} $$

A higher PTR means more students per teacher, while a lower PTR means fewer students per teacher.

The project also uses the existing state-wise `PTR` field. The aggregation method must be stated when reporting averages.

G. Zero-Enrolment School Percentage

This indicator measures the percentage of schools reporting zero student enrolment.

$$ Zero\ Enrolment(\%) = \frac{Zero\ Enrolment\ Schools} {Total\ Schools}\times100 $$

H. Single-Teacher School Percentage

This indicator measures the percentage of schools having only one teacher.

$$ Single\ Teacher(\%) = \frac{Single\ Teacher\ Schools} {Total\ Schools}\times100 $$

Analytical purpose:

These two indicators help examine infrastructure-related challenges and differences in school staffing across states.

Table 6: Derived Educational Feature Catalogue and Mathematical Formulations

Feature family	Derived variable	Mathematical formulation	Analytical purpose
Literacy	Gender_Gap	Male − Female	Gender literacy difference
Enrolment	Girls_Share	Girls / Total × 100	Girls' participation
Enrolment	Gender_Gap	Boys − Girls	Enrolment difference
Infrastructure	Avg_Enrolments	Total Enrolments / Total Schools	Average school size
Infrastructure	Avg_Teachers	Total Teachers / Total Schools	Teacher availability
Infrastructure	PTR	Total Enrolments / Total Teachers	Student-teacher relationship
Infrastructure	Zero_Enrolment_Percentage	Zero schools / Total schools × 100	Zero-enrolment prevalence
Infrastructure	Single_Teacher_Percentage	Single-teacher schools / Total schools × 100	Single-teacher school prevalence

3.4 Comprehensive Descriptive Statistical Analysis

Descriptive statistical analysis is performed to understand the distribution and characteristics of the educational indicators.

The project uses R's statistical functions and the `dplyr` package to summarize the literacy, enrolment and infrastructure datasets.

The analysis includes:

- Total observations.
- Mean.
- Median.
- Standard deviation.

- Minimum.
- Maximum.
- Range.
- State-wise rankings.
- Gender differences.
- Infrastructure totals.

A. Literacy Statistical Analysis

The literacy dataset contains the following major variables:

- Male literacy.
- Female literacy.
- Persons literacy.
- Gender literacy gap.

The following R code calculates descriptive statistics:

```
literacy_summary <- literacy_data %>%
  summarise(
    Male_Mean = mean(Male, na.rm = TRUE),
    Female_Mean = mean(Female, na.rm = TRUE),
    Total_Mean = mean(Total, na.rm = TRUE),

    Male_Median = median(Male, na.rm = TRUE),
    Female_Median = median(Female, na.rm = TRUE),

    Male_SD = sd(Male, na.rm = TRUE),
    Female_SD = sd(Female, na.rm = TRUE)
  )
```

Analytical interpretation:

- Mean literacy provides the arithmetic average across state/UT observations.
- Median literacy identifies the middle observation after sorting.
- Standard deviation describes the spread of literacy values.
- Gender gap identifies the difference between male and female literacy.

These measures help identify variations in literacy across Indian states and union territories.

B. Enrolment Statistical Analysis

The enrolment dataset is used to calculate descriptive statistics for boys, girls and total enrolment.

```
enrolment_summary <- enrolment_data %>%
  summarise(
    Total_Boys = sum(Boys, na.rm = TRUE),
    Total_Girls = sum(Girls, na.rm = TRUE),
    Total_Students = sum(Total, na.rm = TRUE),

    Average_Boys = mean(Boys, na.rm = TRUE),
    Average_Girls = mean(Girls, na.rm = TRUE),

    Median_Total = median(Total, na.rm = TRUE)
  )
```

Analytical interpretation:

- Total boys' enrolment measures the number of enrolled boys represented by the dataset.
- Total girls' enrolment measures the number of enrolled girls.
- Total enrolment provides the combined student count.
- Average and median enrolment help understand differences in state-level enrolment.

C. Infrastructure Statistical Analysis

The infrastructure dataset is used to understand the distribution of schools, students and teachers.

The project's previously calculated summary is:

Table 7: Verified Infrastructure Summary for UDISE 2023–24

Indicator	Project summary
States and Union Territories	36
Total schools	1,471,891
Total students	248,045,828
Total teachers	9,807,600
Average state/UT PTR	21.2

The PTR value of 21.2 is the unweighted average of state/UT PTR values, not the ratio of total students to total teachers.

R implementation:

```
infrastructure_summary <- infrastructure_data %>%
  summarise(
    Total_States = n_distinct(State),

    Total_Schools =
      sum(Total_Schools, na.rm = TRUE),

    Total_Students =
      sum(Total_Enrolments, na.rm = TRUE),

    Total_Teachers =
      sum(Total_Teachers, na.rm = TRUE),

    Average_PTR =
      mean(PTR, na.rm = TRUE)
  )
```

Analytical interpretation:

The infrastructure summary provides an overview of the educational system represented in the 2023–24 dataset. However, total school and student counts are influenced by state population and size. Therefore, indicators such as PTR, average teachers per school and single-teacher school percentage provide additional

3.5 State-Wise Comparative Analysis

State-wise comparative analysis is used to identify variations between Indian states and union territories.

The project performs comparisons across three major dimensions.

1. Literacy comparison

- Male literacy by state.
- Female literacy by state.
- Persons literacy by state.
- Gender literacy gap by state.

2. Enrolment comparison

- Boys' enrolment by state.

- Girls' enrolment by state.
- Total enrolment by state.
- Girls' enrolment share by state.

3. Infrastructure comparison

- Total schools by state.
- Total students by state.
- Total teachers by state.
- Pupil-teacher ratio by state.
- Zero-enrolment school percentage.
- Single-teacher school percentage.

The following R operations support these comparisons:

```
literacy_data %>%
  arrange(desc(Total))
enrolment_data %>%
  arrange(desc(Total))
infrastructure_data %>%
  arrange(desc(PTR))
```

These operations arrange the observations in descending order and help identify the highest values for the selected indicator.

3.6 Correlation and Relationship Analysis

Correlation analysis helps investigate whether two numeric educational indicators move together.

For example, the project can investigate:

- Male literacy versus female literacy.
- Total schools versus total enrolment.
- Total teachers versus total enrolment.
- PTR versus average teachers per school.
- Girls' enrolment versus boys' enrolment.

Pearson's correlation coefficient is calculated using:

$$ r = \frac{\sum (x_i-\bar{x})(y_i-\bar{y})} {\sqrt{\sum(x_i-\bar{x})^2\sum(y_i-\bar{y})^2}} $$

The coefficient ranges between -1 and +1.

Correlation value	Interpretation
Near +1	Strong positive linear relationship
Near 0	Weak linear relationship
Near -1	Strong negative linear relationship

R implementation:

```
cor(
  literacy_data$Male,
  literacy_data$Female,
  use = "complete.obs",
  method = "pearson"
)
```

For infrastructure:

```
cor(
  infrastructure_data$Total_Enrolments,
  infrastructure_data$Total_Teachers,
  use = "complete.obs",
  method = "pearson"
)
```

Analytical purpose:

Correlation analysis helps identify relationships between educational indicators. However, correlation does not establish causation. For example, a positive relationship between total schools and total enrolment may partly reflect differences in state population.

3.7 Advanced Exploratory Visualization and Analytical Techniques

The project uses different visualization techniques to examine the educational datasets.

A. Literacy Scatter Plot

The scatter plot compares male and female literacy rates.

- X-axis: Male literacy.
- Y-axis: Female literacy.
- Each point represents a state/UT.
- A diagonal reference line represents equal male and female literacy.

Analytical purpose: To understand the relationship between male and female literacy and identify states with larger gender differences.

B. Enrolment Line and Bar Charts

These charts compare boys' and girls' enrolment across states or educational categories.

Analytical purpose:

- Examine enrolment differences.
- Identify states with higher or lower enrolment.
- Compare boys' and girls' participation.

A bar chart is generally more suitable for unordered state categories. A line chart is more suitable when the horizontal axis represents a meaningful sequence, such as time or educational stage.

C. Literacy Heat Map

The heat map represents state-wise literacy indicators using colour intensity.

Possible columns include:

- Male literacy.
- Female literacy.
- Persons literacy.
- Gender literacy gap.

Analytical purpose: To identify patterns and differences in literacy indicators across states.

The gender gap is expressed in percentage points, whereas literacy rates are percentages. The heat-map scale must account for this difference.

D. Gender Enrolment Donut Chart

The donut chart presents the composition of boys' and girls' enrolment.

\[Boys\ Share = \frac{Boys}{Total}\times100 \]

\[Girls\ Share = \frac{Girls}{Total}\times100 \]

Analytical purpose: To communicate the proportion of boys and girls in the selected enrolment dataset.

E. Bollinger Band Visualization

The Bollinger Band chart is included as an advanced visualization practical.

It uses:

- Rolling mean.
- Rolling standard deviation.
- Upper band.
- Lower band.
- A shaded volatility region.

The mathematical formulation is:

$$Middle\ Band = MA_n(t)$$

$$Upper\ Band = MA_n(t)+2SD_n(t)$$

$$Lower\ Band = MA_n(t)-2SD_n(t)$$

Important: Your earlier Bollinger Band exercise used simulated sample values. It demonstrates a visualization technique and is not a real educational indicator or an education-related statistical finding.

3.9 EDA Summary and Visualization Outputs

The following figures can be included in your report under the EDA chapter.

Figure	Visualization	Data source
Figure 2	Modular R analytical workflow	Project architecture
Figure 3	Male vs female literacy scatter plot	Census 2011
Figure 4	State-wise literacy bar chart	Census 2011
Figure 5	Boys vs girls enrolment chart	UDISE 2021–22
Figure 6	Literacy heat map	Census 2011
Figure 7	Gender enrolment donut chart	UDISE 2021–22
Figure 8	Schools by state	UDISE 2023–24

Figure 9 Pupil-teacher ratio by state UDISE 2023–24

Figure 10 Zero and single-teacher school comparison UDISE 2023–24

Figure 11 Bollinger Band demonstration Simulated sample data

Overall EDA conclusion

The EDA process transforms the raw educational datasets into structured analytical information. Through data cleaning, feature engineering, descriptive statistics, state-wise comparisons and visualization, the project examines educational differences across India.

The analysis provides a foundation for the interactive dashboard, where users can explore literacy, enrolment, infrastructure and gender equality through charts, tables and state-level filters.

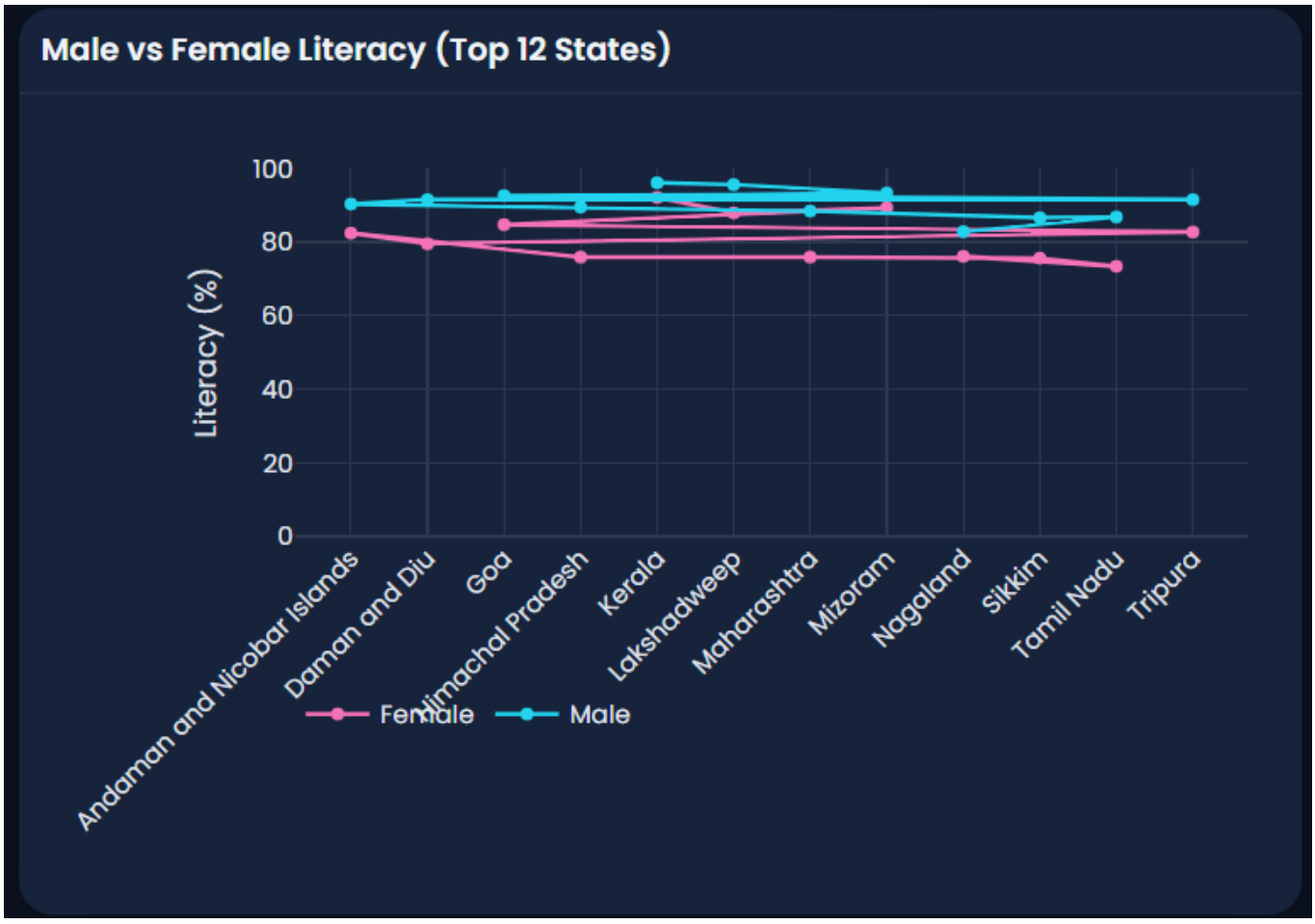


Figure 3 Male vs female literacy scatter plot Census 2011

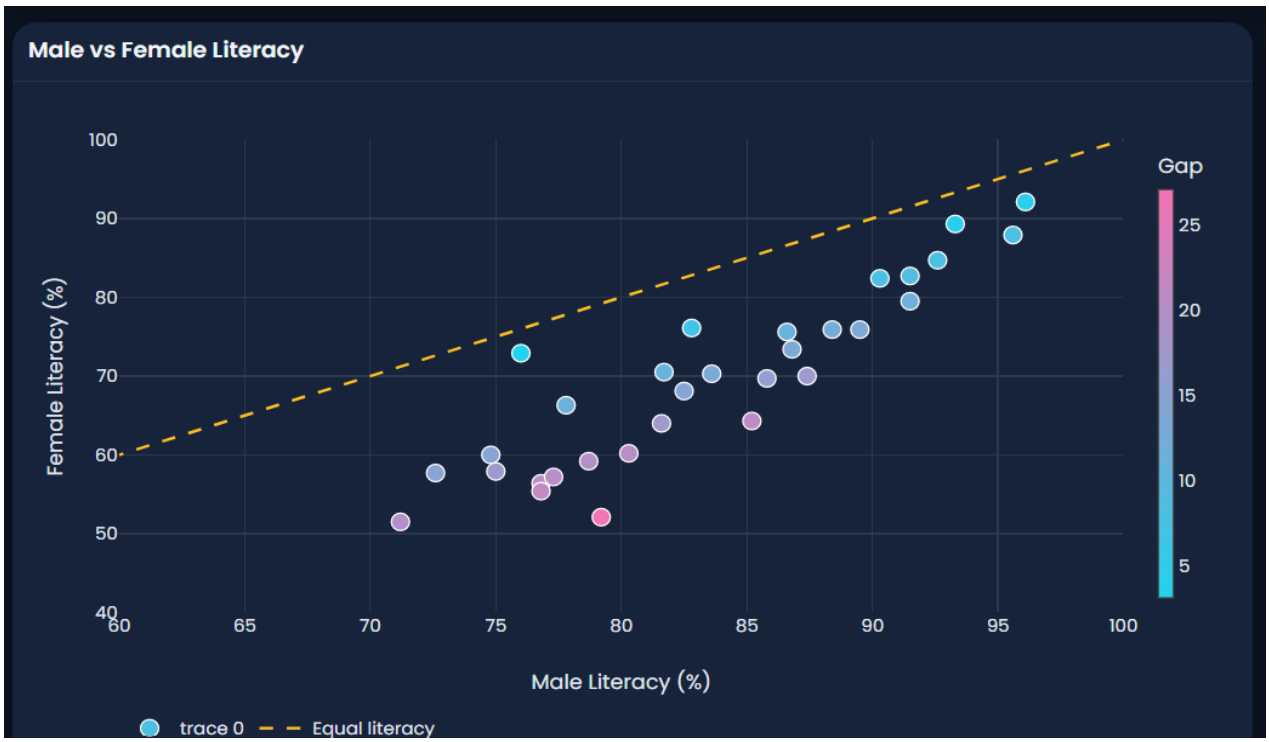


Figure 3. Male vs female literacy scatter plot Census 2011

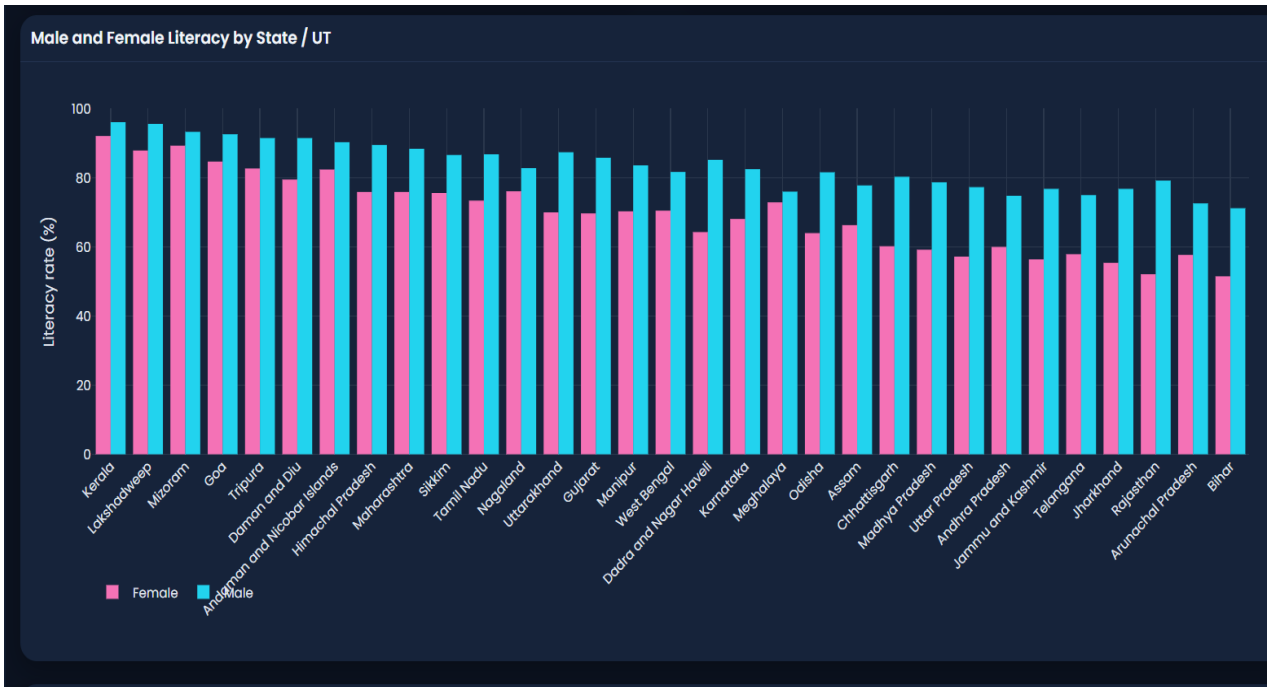


Figure 4. State-wise literacy bar chart Census 2011

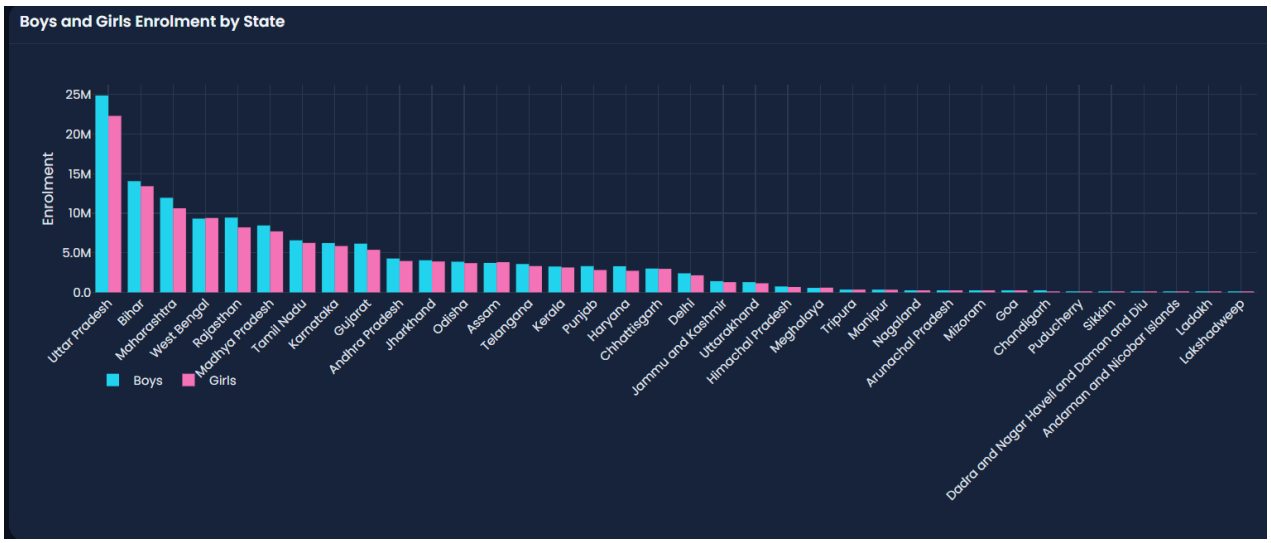


Figure 5. Boys vs girls enrolment chart UDISE 2021–22.

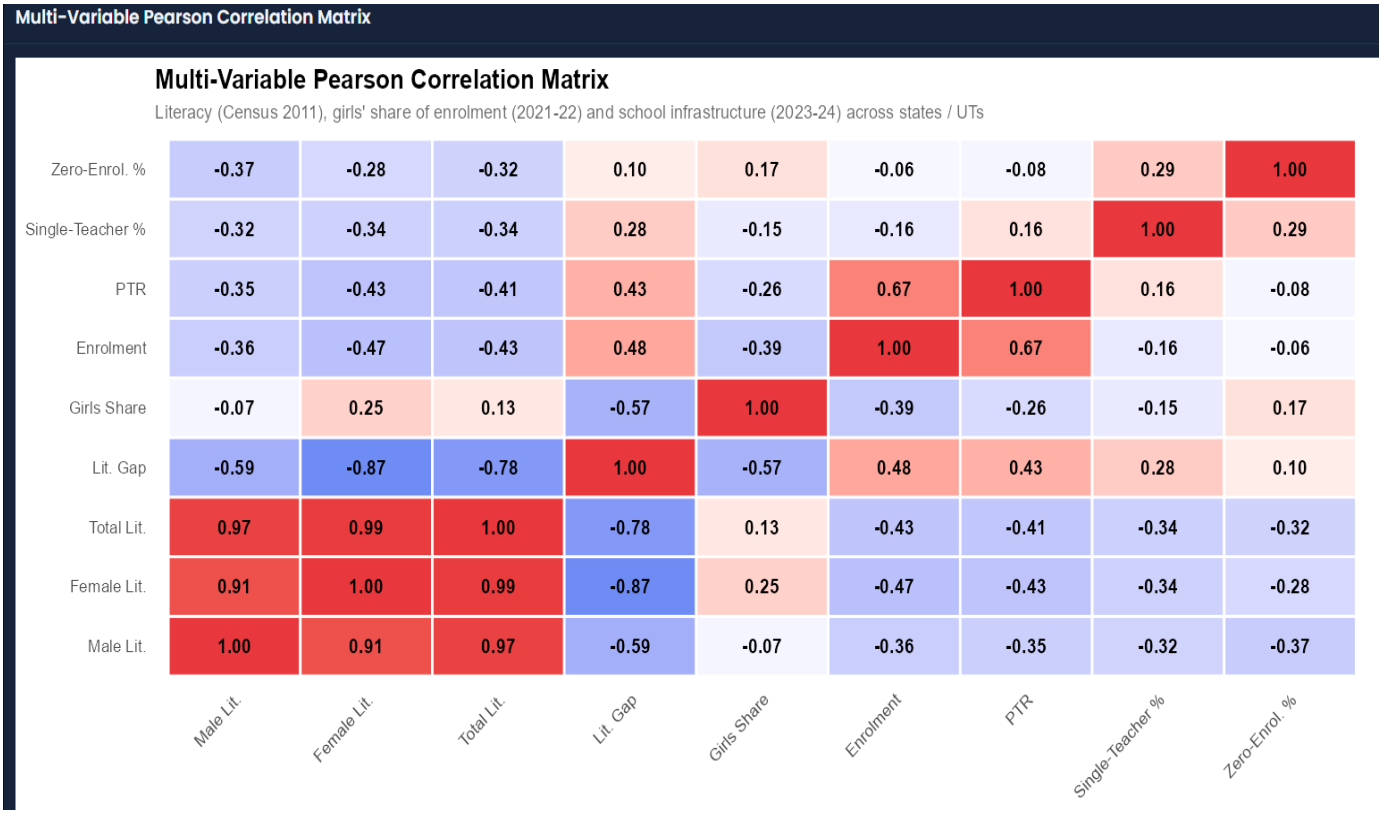


Figure 6. Literacy heat map Census 2011.

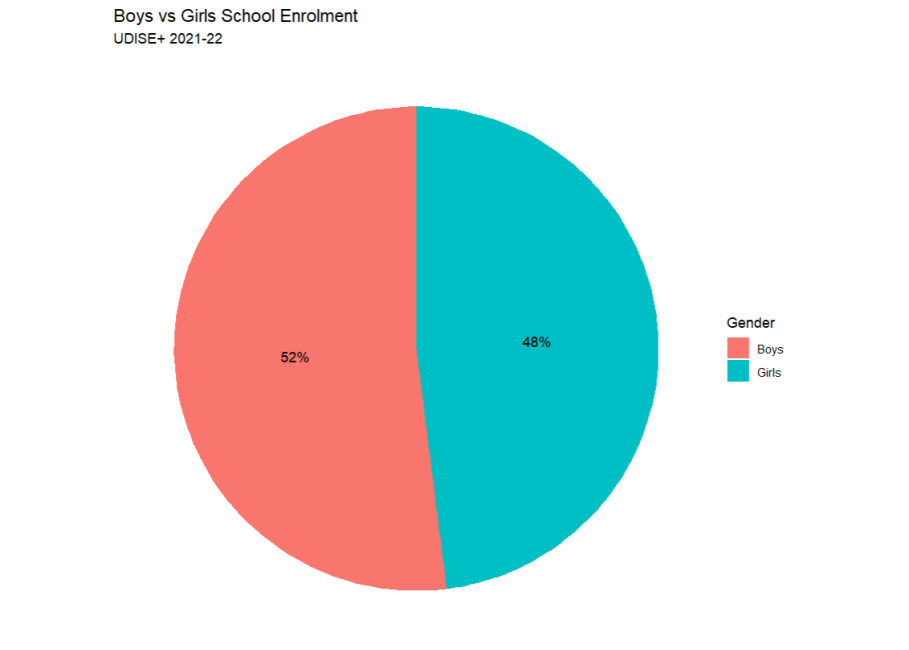


Figure 7. Gender enrolment donut chart UDISE 2021–22.

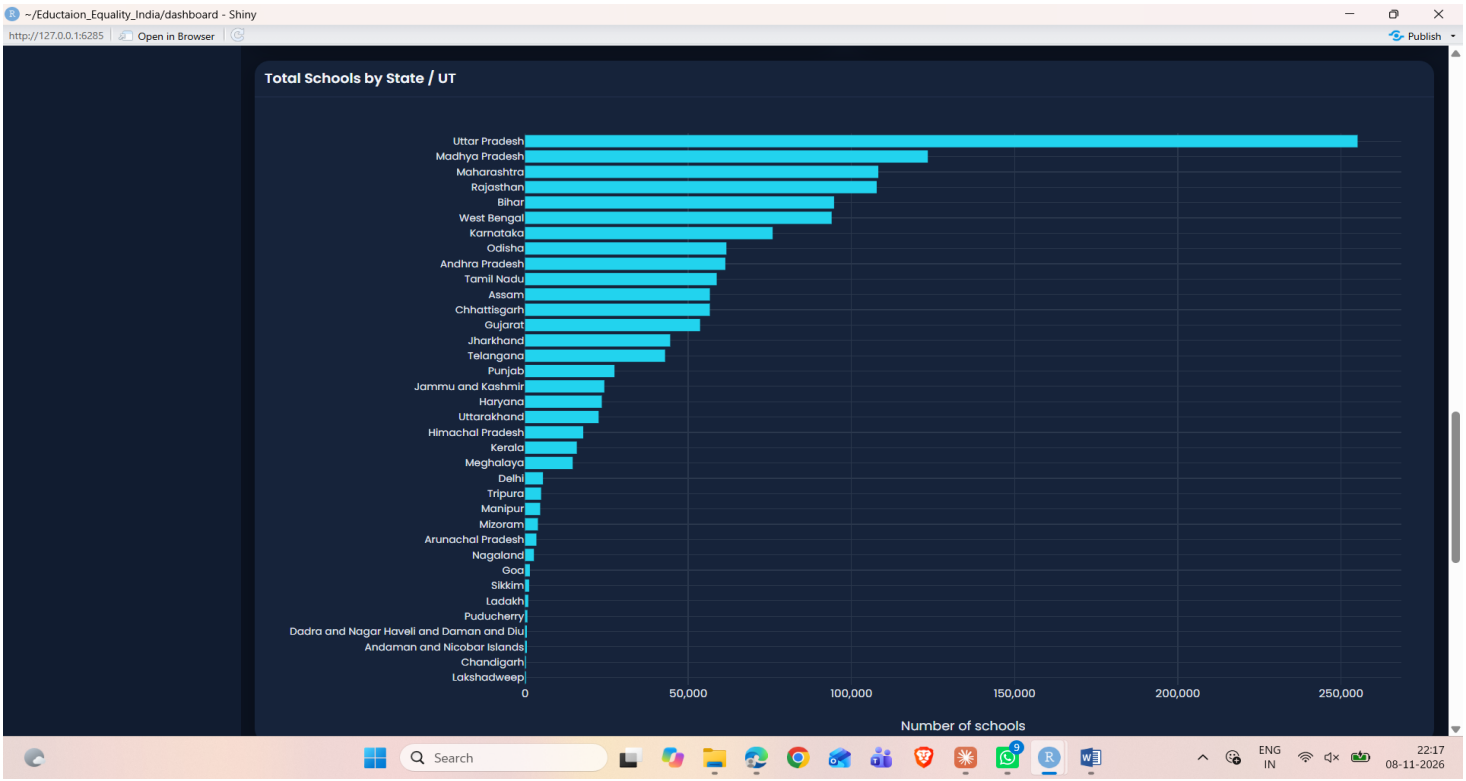


Figure 8. Schools by state UDISE 2023–24

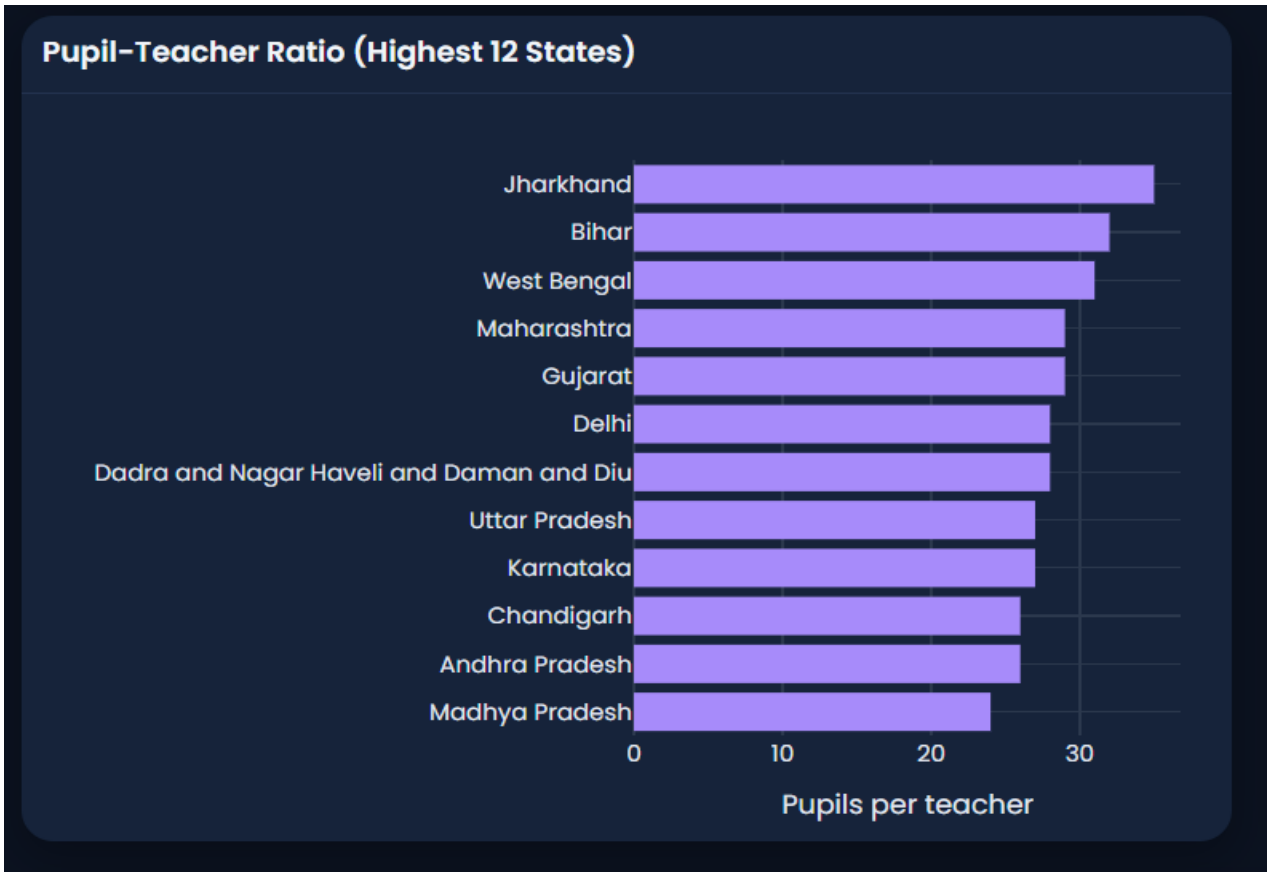


Figure 9. Pupil-teacher ratio by state UDISE 2023–24.

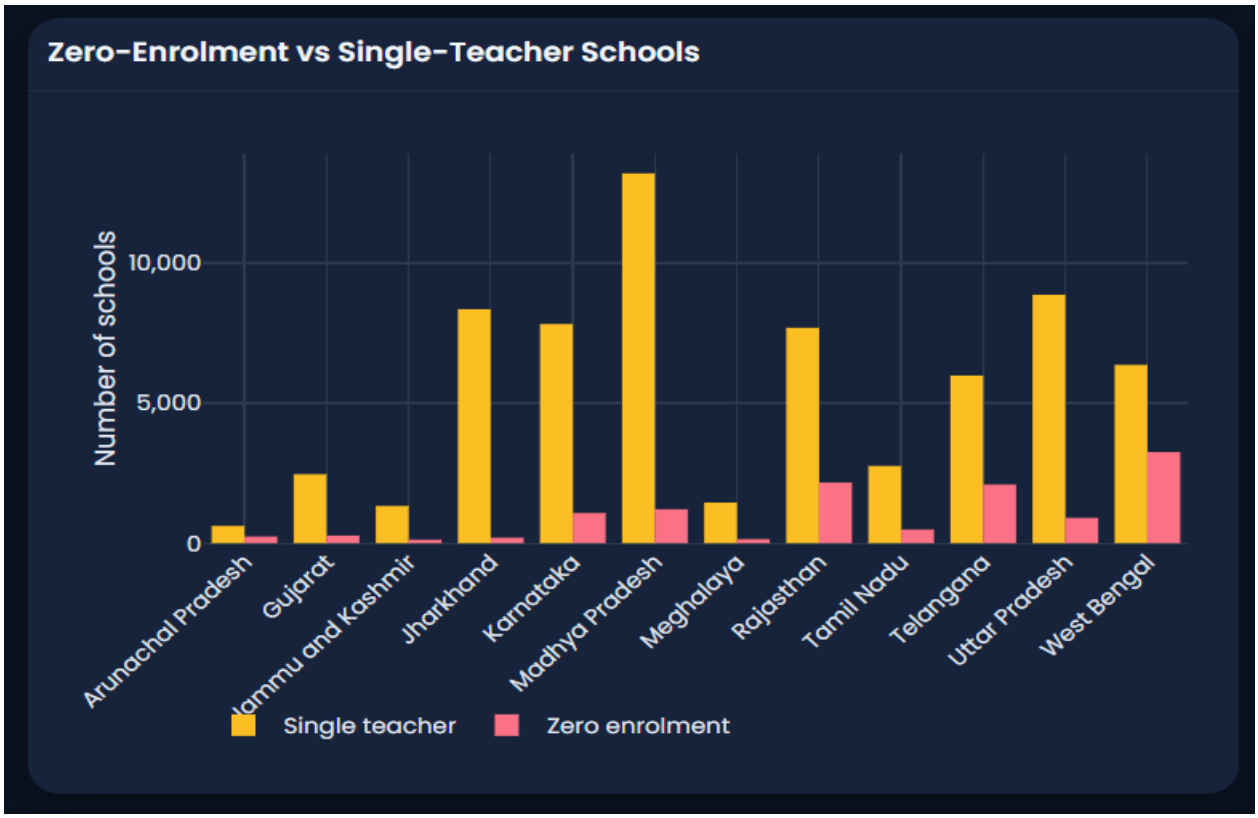


Figure 10 .Zero and single-teacher school comparison UDISE 2023–24.

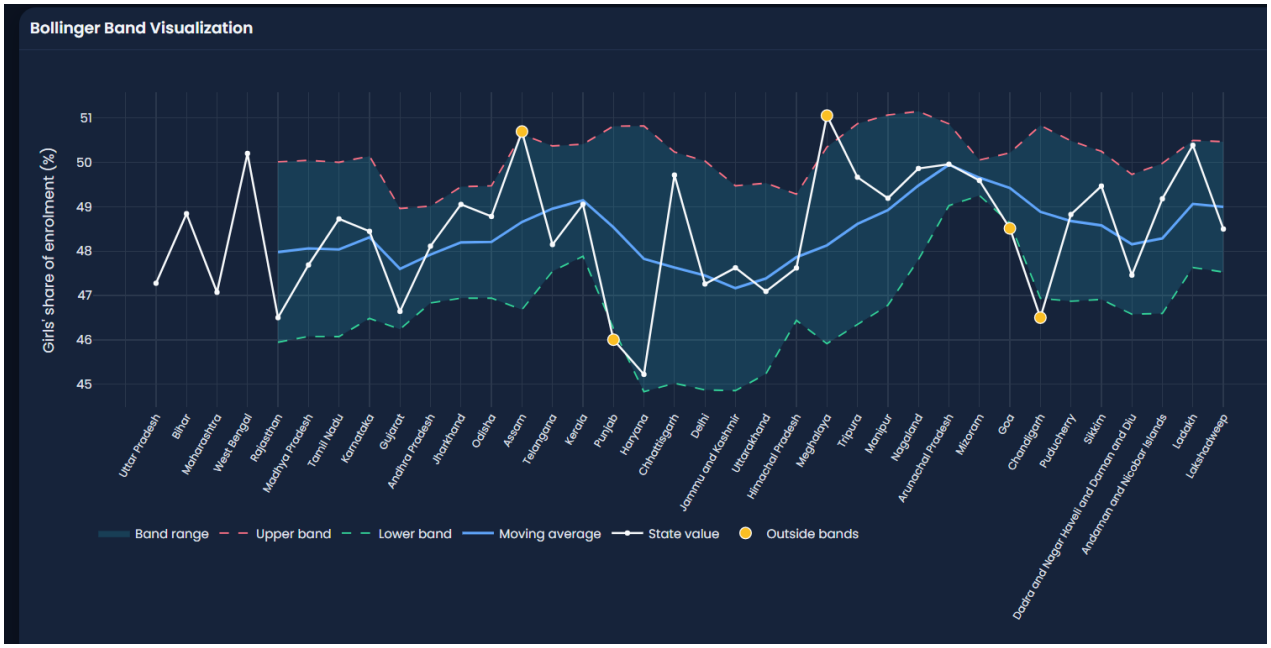


Figure 11. Bollinger Band demonstration n Simulated sample data.

4. ANALYSIS ON DATASET

Each analytical module follows the five-part reporting structure used in the reference report: introduction, general description, specific requirements/functions/formulas, analysis results and visualisation. Results are reported only where the supplied data or captured project charts support them.

4.1 Analysis 1: Boys versus Girls Enrolment Composition

i. Introduction

Gender-wise enrolment composition provides an initial view of participation in school education. The comparison asks whether the total number of reported boys and girls enrolled is broadly balanced in the UDISE+ 2021–22 all-ages, Pre-Primary to Higher Secondary table.

ii. General Description

The analysis sums boys' and girls' enrolment across State/UT rows after excluding the India aggregate. The total composition is displayed as a pie chart. A separate stage-wise grouped bar chart compares the counts across Pre-Primary, Primary, Upper Primary, Secondary and Higher Secondary.

iii. Specific Requirements, Functions and Formulas

```
boys_total  <- sum(enrolment_data$Boys, na.rm = TRUE)
girls_total <- sum(enrolment_data$Girls, na.rm = TRUE)

boys_share  <- boys_total / (boys_total + girls_total) * 100
girls_share <- girls_total / (boys_total + girls_total) * 100
```

iv. Analysis Results

The dashboard pie chart reports a composition of approximately 52% boys and 48% girls. The small difference indicates that the total reported enrolment is relatively close to parity at the aggregate level, but an aggregate split can conceal State-level and school-stage variation.

3.8 EDA Results and Analytical Interpretation

The exploratory analysis provides three main perspectives on education equality in India.

Analysis area	Dataset	Main analytical objective
Literacy	Census 2011	Compare male, female and persons literacy
Enrolment	UDISE 2021–22	Compare boys' and girls' enrolment
Infrastructure	UDISE 2023–24	Examine schools, students, teachers and PTR

The infrastructure summary confirms the scale of the dataset, covering 36 states/UTs and more than 1.47 million schools.

The literacy dataset enables examination of historical gender literacy differences, while the enrolment dataset provides a separate view of student participation.

The infrastructure dataset adds information about the availability and distribution of schools and teachers.

Together, these analyses provide a multidimensional descriptive view of education equality. However, because the datasets refer to different years, their results should not be interpreted as simultaneous measurements.

v. Visualization

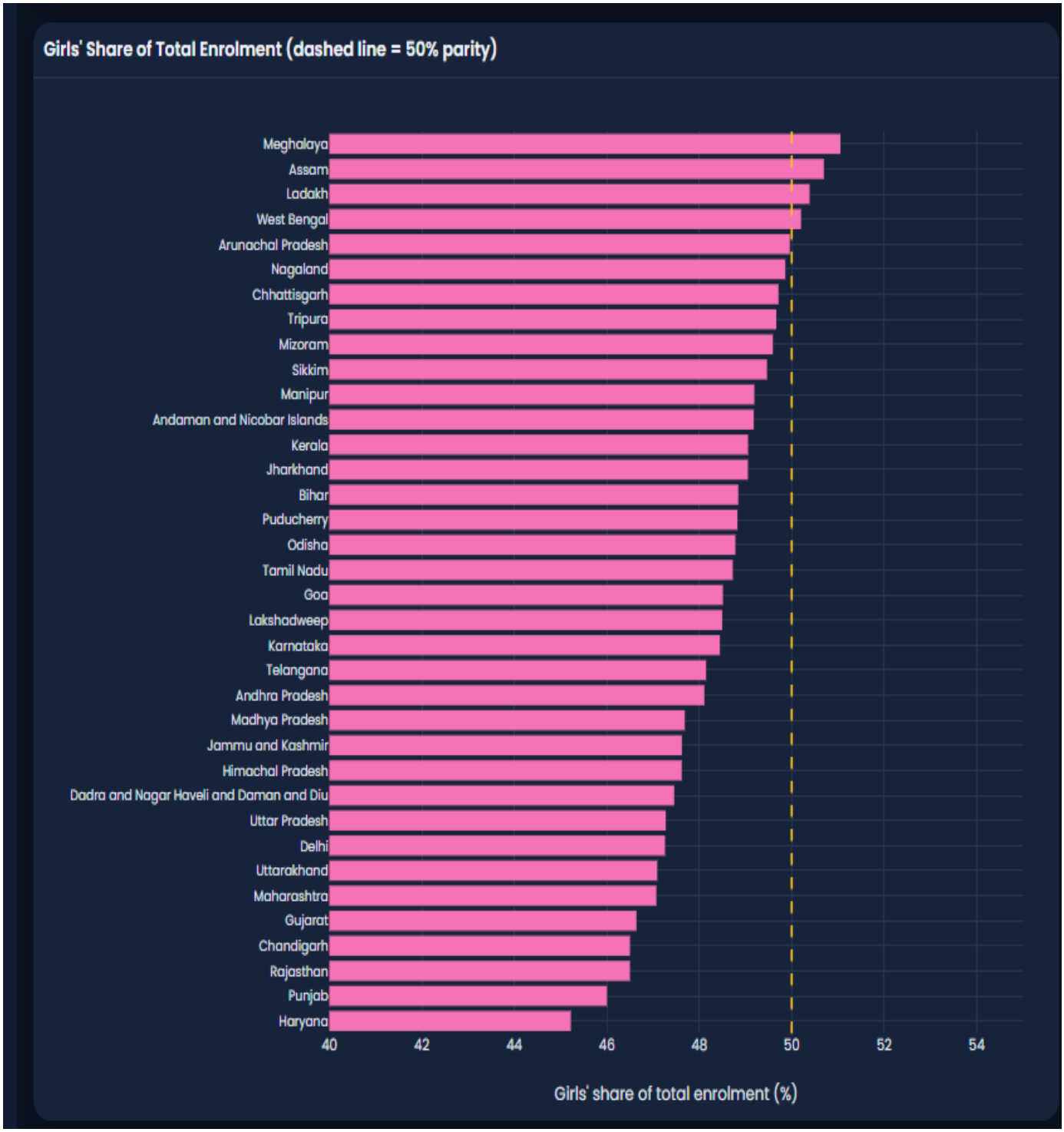


Figure 1: Boys versus girls school enrolment composition, UDISE+ 2021–22.4.2 Analysis 2: State-wise Literacy Heat Map

i. Introduction

A heat map is useful when several numeric indicators must be compared across many States/UTs. Colour intensity helps identify broad patterns and outlying values without requiring the reader to inspect every row of a table.

ii. General Description

The Census 2011 literacy table is transformed into a long format with indicators for female literacy, gender gap, male literacy and persons literacy. Each row represents a State/UT and each column represents an indicator.

iii. Specific Requirements, Functions and Formulas

```
literacy_long <- literacy_data %>%
  select(State, Female, Gender_Gap_pp, Male, Total) %>%
  pivot_longer(-State, names_to = "Indicator",
               values_to = "Percentage")

ggplot(literacy_long,
       aes(x = Indicator, y = State, fill = Percentage)) +
  geom_tile() +
  scale_fill_viridis_c() +
  labs(title = "State-wise Literacy Heat Map",
       x = "Literacy Indicator", y = "State / UT")
```

iv. Analysis Results

The supplied Census table contains 31 State/UT records after removing the national aggregate. The female literacy rate is highest in Kerala (92.1%) and lowest in Bihar (51.5%). The heat map also shows that the gender gap is not uniform across the country.

v. Visualization

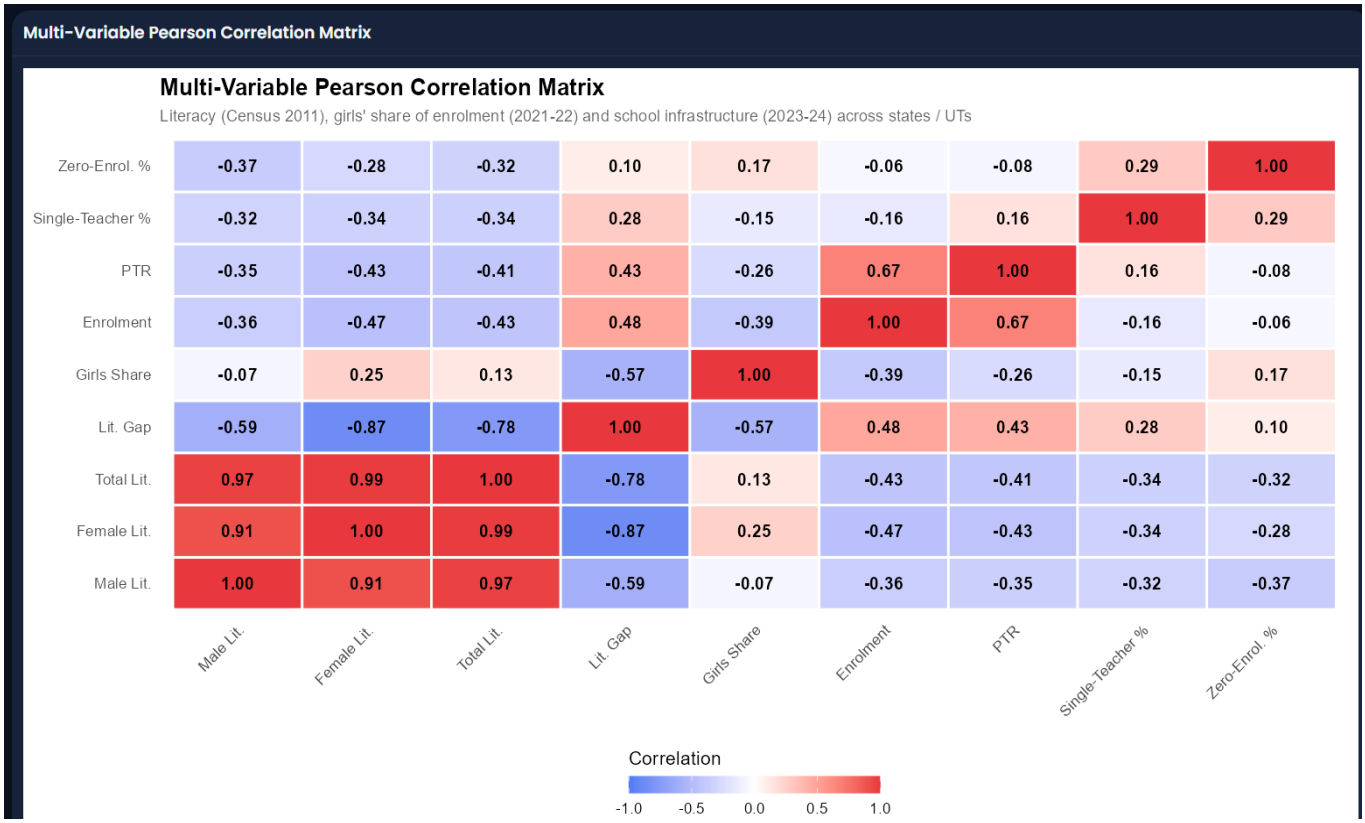


Figure 2: State-wise literacy heat map based on Census 2011.

4.3 Analysis 3: Male–Female Literacy Gap

i. Introduction

A literacy gap is a direct descriptive measure of gender difference. The gap is calculated as male literacy minus female literacy. A positive value indicates a higher male literacy rate in the reported State/UT.

ii. General Description

The project uses a dumbbell chart to display male and female literacy rates on a common horizontal axis. Each State/UT has two endpoints joined by a line, making the magnitude and direction of the difference visible.

iii. Specific Requirements, Functions and Formulas

```
literacy_data <- literacy_data %>%
  mutate(Gender_Gap_pp = Male - Female) %>%
  arrange(desc(Gender_Gap_pp))

# Gender_Gap_pp is measured in percentage points.
```

iv. Analysis Results

The unweighted average of State/UT literacy gaps is 13.9 percentage points. The chart shows larger gaps for several lower-literacy States/UTs and smaller gaps in some higher-literacy regions. This is a descriptive pattern and does not explain the causes of the differences.

v. Visualization

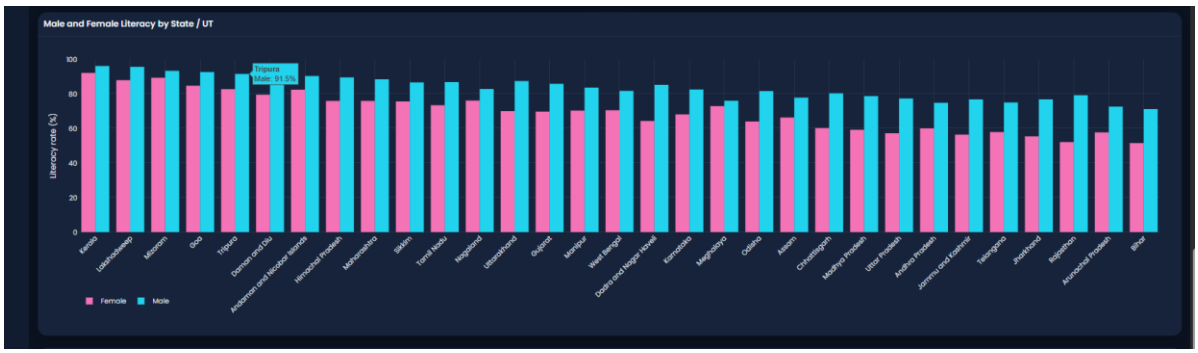


Figure 4: Male–female literacy gap by State/UT (dumbbell chart).

4.4 Analysis 4: Distribution of School Infrastructure

i. Introduction

The number of schools is a basic measure of the scale of the formal school network. It is useful for understanding the distribution of institutions, but school counts alone do not measure quality, accessibility or adequacy relative to population.

ii. General Description

The UDISE+ 2023–24 infrastructure file provides school, enrolment and teacher counts for each State/UT. The top ten States/UTs by school count are sorted in descending order and presented as a horizontal bar chart.

iii. Specific Requirements, Functions and Formulas

```
top_schools <- infrastructure_data %>%
  arrange(desc(Total_Schools)) %>%
  slice_head(n = 10)

ggplot(top_schools,
       aes(x = reorder(State, Total_Schools),
           y = Total_Schools)) +
  geom_col() + coord_flip() +
  labs(x = "State / UT", y = "Number of schools")
```

iv. Analysis Results

The State/UT with the largest school count in the supplied table is Uttar Pradesh (255,087 schools). Across all 36 State/UT records, the total is 1,471,891 schools. These are counts, so larger population and geographic size can contribute to higher totals.

v. Visualization

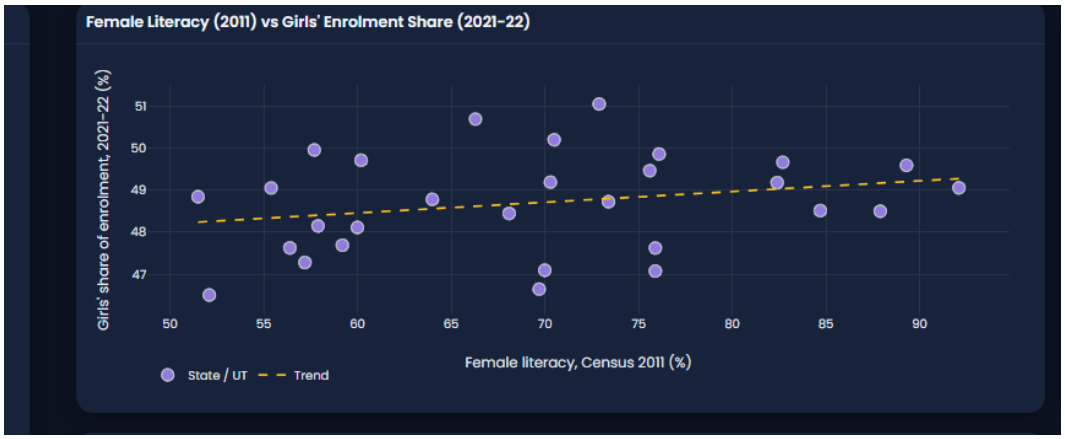


Figure 5: Top 10 States/UTs by number of schools, UDISE+ 2023–24.

4.5 Analysis 5: Enrolment by School Stage

i. Introduction

Enrolment varies across school stages. A stage-wise comparison helps identify where the largest number of students is recorded and whether the gender composition changes between stages.

ii. General Description

The UDISE+ 2021–22 source includes separate boys, girls and total columns for Pre-Primary, Primary, Upper Primary, Secondary and Higher Secondary stages. The grouped bar chart compares boys and girls within each stage.

iii. Specific Requirements, Functions and Formulas

```
stage_long <- stage_data %>%
  pivot_longer(c(Boys, Girls),
               names_to = "Gender",
               values_to = "Enrolment")

ggplot(stage_long,
       aes(x = Stage, y = Enrolment, fill = Gender)) +
  geom_col(position = "dodge")
```

iv. Analysis Results

The chart shows the largest enrolment at the Primary stage, followed by Upper Primary, Secondary and Higher Secondary. Boys' counts are higher than girls' counts in the displayed stages. The chart is a count comparison, not a transition or retention rate; it cannot by itself establish how many students progress from one stage to the next.

v. Visualization

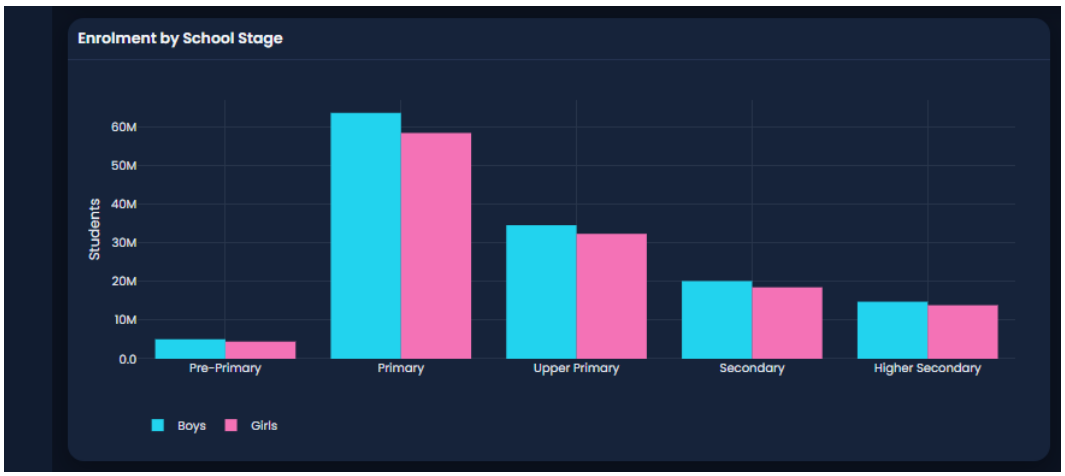


Figure 6: Enrolment by school stage and gender, UDISE+ 2021–22.

4.6 Analysis 6: Zero-Enrolment and Single-Teacher Schools

i. Introduction

School functionality and staffing are important dimensions of educational opportunity. Schools reporting zero enrolment and schools with a single teacher may require contextual examination by education administrators.

ii. General Description

The infrastructure table contains counts of schools with zero enrolments and schools with single teachers. A grouped bar chart compares these counts for selected States/UTs.

iii. Specific Requirements, Functions and Formulas

```
infra_selected <- infrastructure_data %>%
  select(State, Zero_Enrolment_Schools,
         Single_Teacher_Schools) %>%
  pivot_longer(-State, names_to = "Indicator",
               values_to = "School_Count")
```

iv. Analysis Results

The displayed chart shows substantial variation across States/UTs. A high count may reflect a larger school network as well as a higher incidence. To compare the relative scale fairly, counts should be converted into percentages of all schools in each State/UT; the count chart alone should not be interpreted as a rate.

v. Visualization

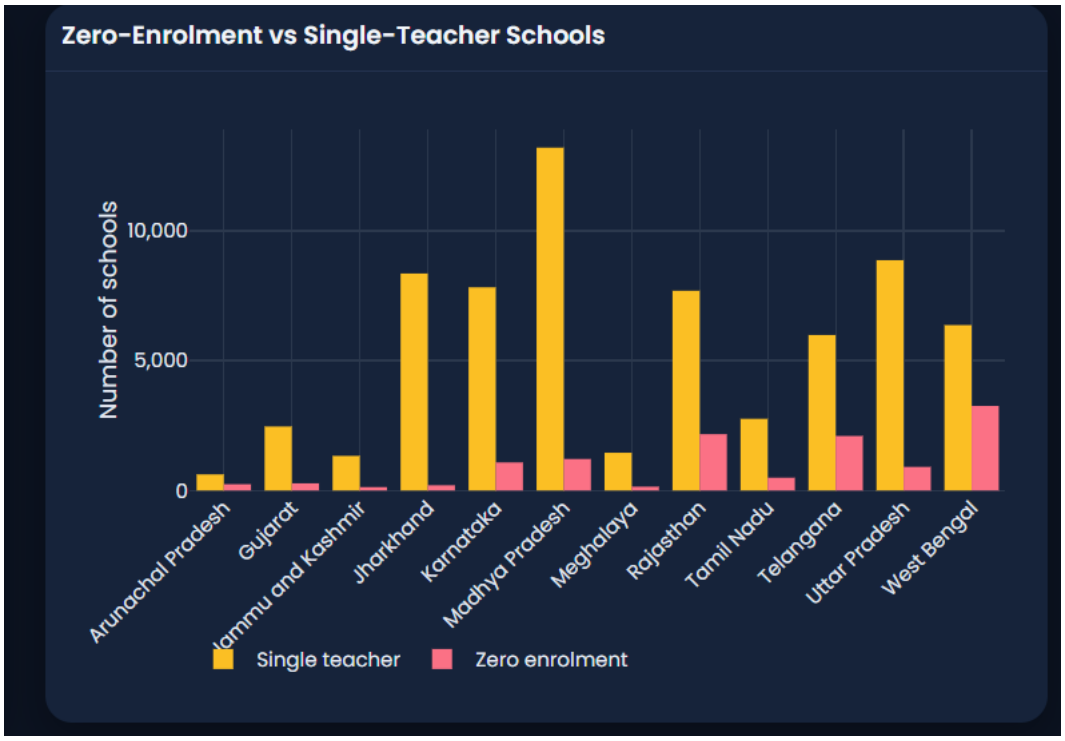


Figure 7: Zero-enrolment versus single-teacher school counts.

4.7 Analysis 7: Literacy Relationship and Distribution

i. Introduction

Scatter plots and boxplots provide complementary views of literacy. The scatter plot compares male and female rates for each State/UT; the boxplot summarises the distribution of the three literacy measures across the State/UT observations.

ii. General Description

The equal-literacy reference line (y = x) marks where male and female literacy would be equal. Points below the line indicate higher male literacy than female literacy. The boxplot compares median, spread and possible outlying State/UT observations.

iii. Specific Requirements, Functions and Formulas

```
ggplot(literacy_data, aes(Male, Female, text = State)) +
  geom_point() +
  geom_abline(intercept = 0, slope = 1,
              linetype = "dashed") +
  labs(x = "Male literacy (%)",
       y = "Female literacy (%)")

# Convert the ggplot object to Plotly:
plotly::ggplotly(p)
```

iv. Analysis Results

The plotted points lie predominantly below the equal-literacy line, indicating that male literacy exceeds female literacy in most displayed State/UT observations. The distribution chart shows a wider spread in female literacy than in male literacy, suggesting greater cross-State variation in the female series.

v. Visualization

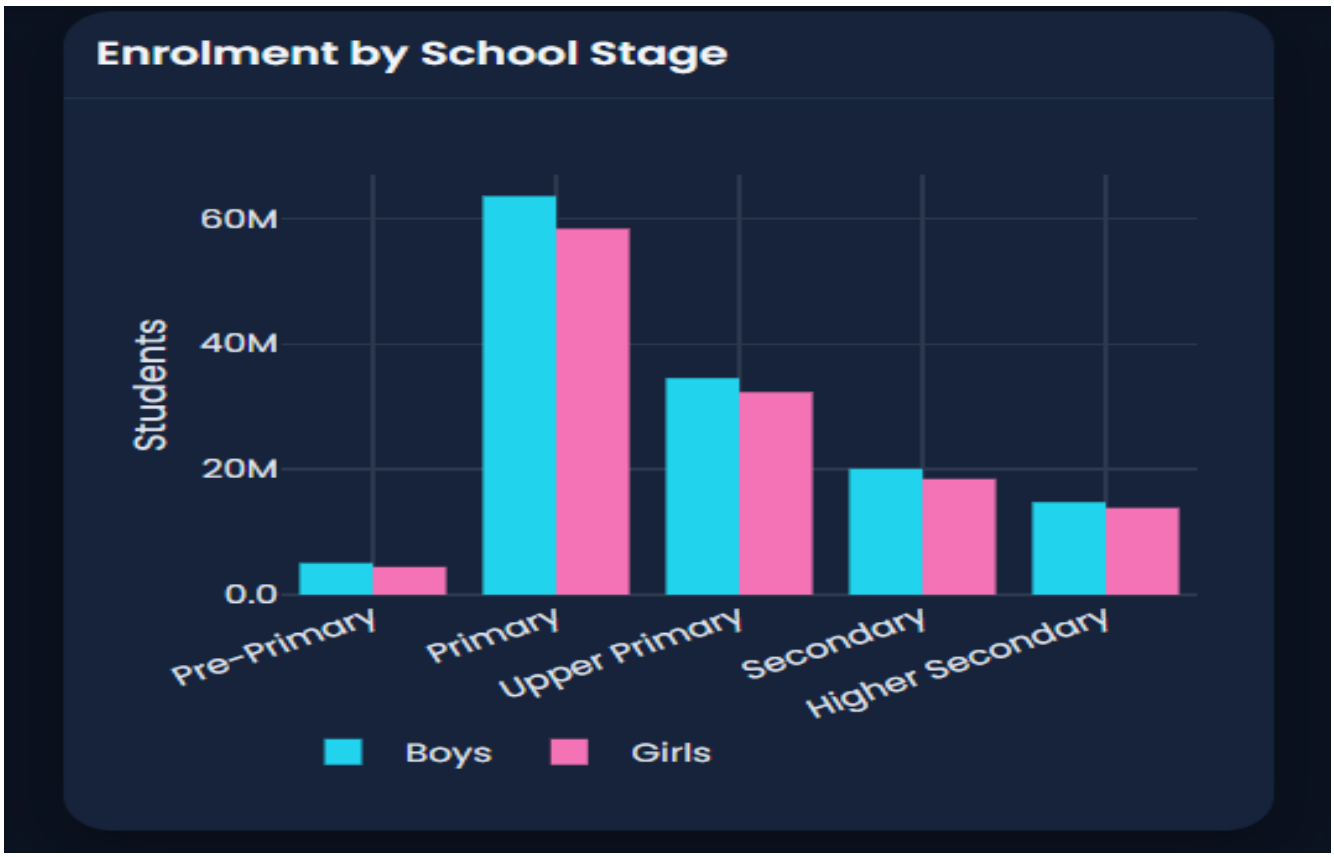


Figure 8: Male versus female literacy scatter plot.

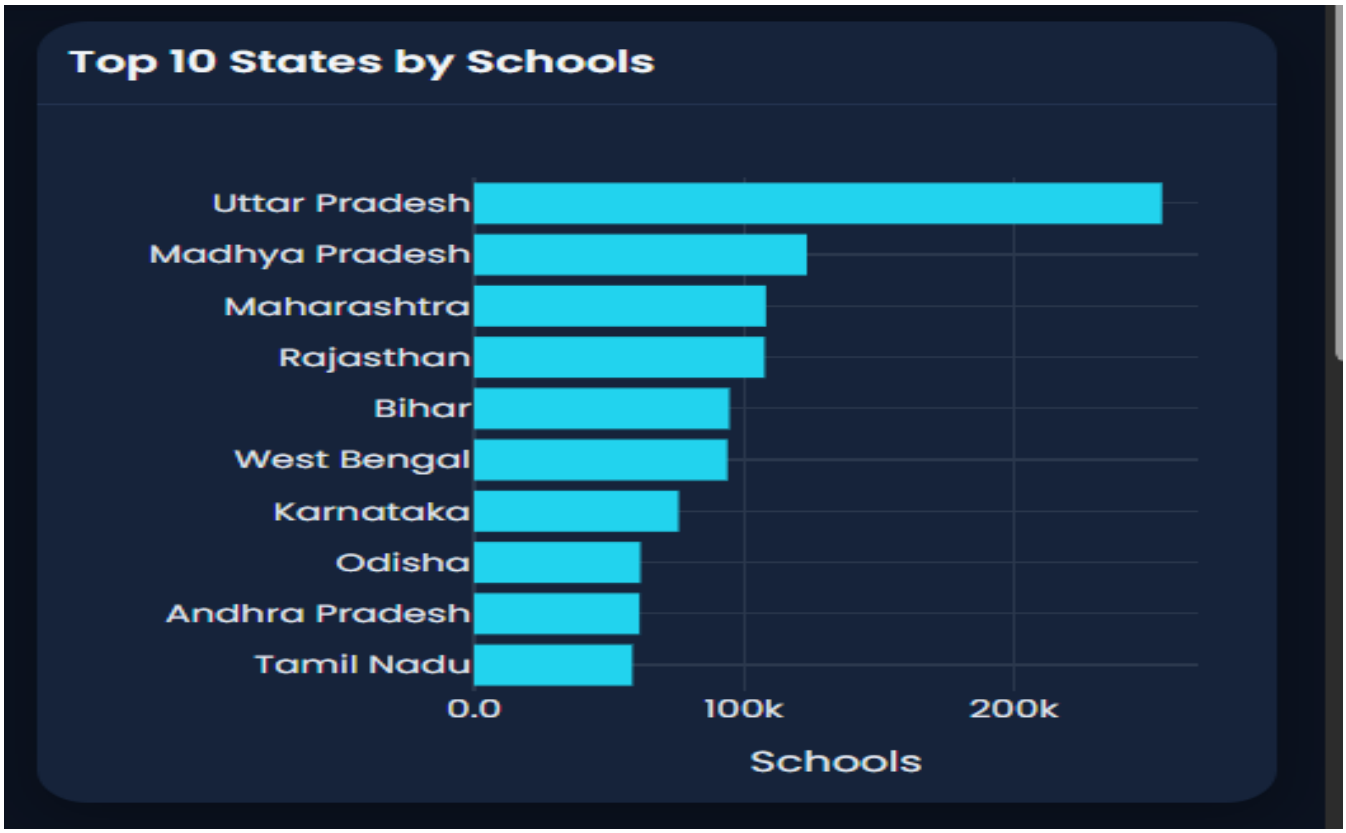


Figure 9: Distribution of male, female and persons literacy rates.

4.8 Analysis 8: Pupil–Teacher Ratio

i. Introduction

The pupil–teacher ratio (PTR) is a commonly used indicator of the relationship between reported pupil numbers and teacher numbers. A higher ratio indicates more pupils per teacher in the reported measure, although it does not capture class size, teacher attendance, subject specialisation or school-level distribution.

ii. General Description

The project ranks States/UTs by the PTR field in UDISE+ 2023–24 and plots the highest values. The table also contains average teachers per school and average enrolments per school, which may be used for further contextual comparison.

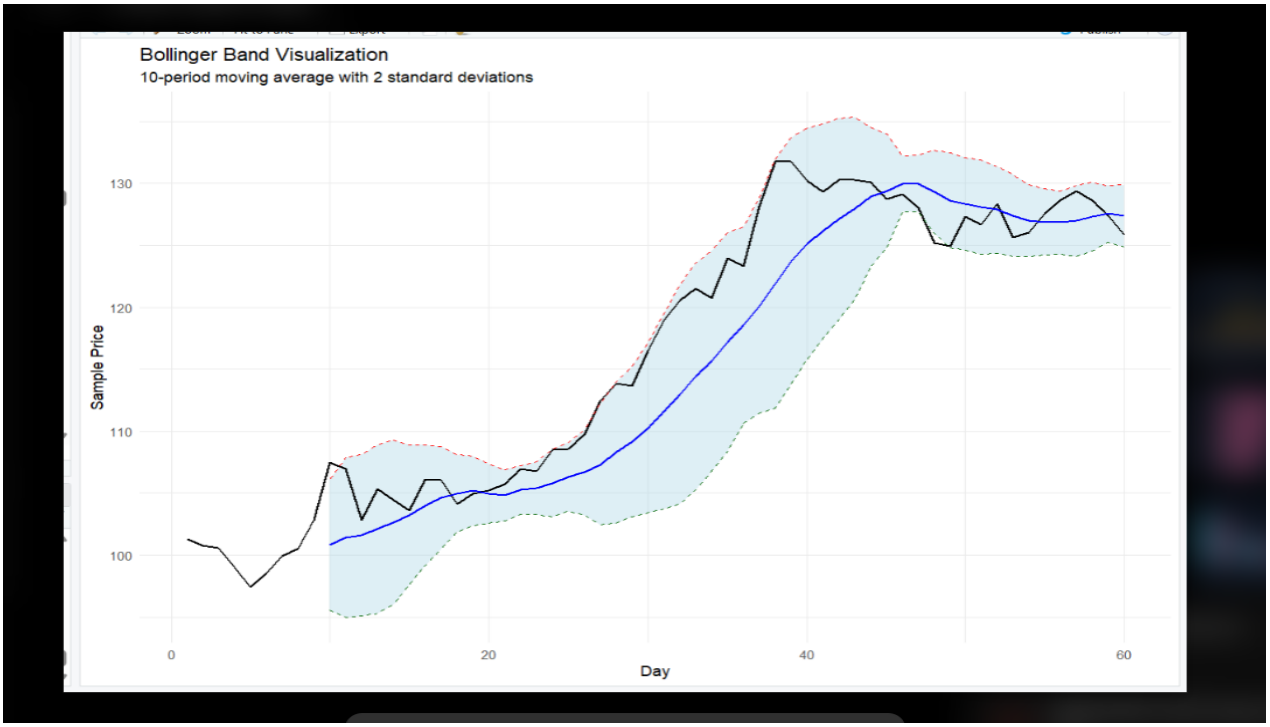
iii. Specific Requirements, Functions and Formulas

```
ptr_rank <- infrastructure_data %>%
  arrange(desc(PTR)) %>%
  slice_head(n = 12)

# Unweighted mean of state ratios:
mean(infrastructure_data$PTR, na.rm = TRUE)
```

iv. Analysis Results

The highest PTR in the supplied file is recorded for Jharkhand (35.0 pupils per teacher). The unweighted mean of State/UT PTR values is 21.2. This mean is not a national weighted PTR; a weighted ratio w ould require the appropriate total pupil and teacher denominators.



v. Visualization

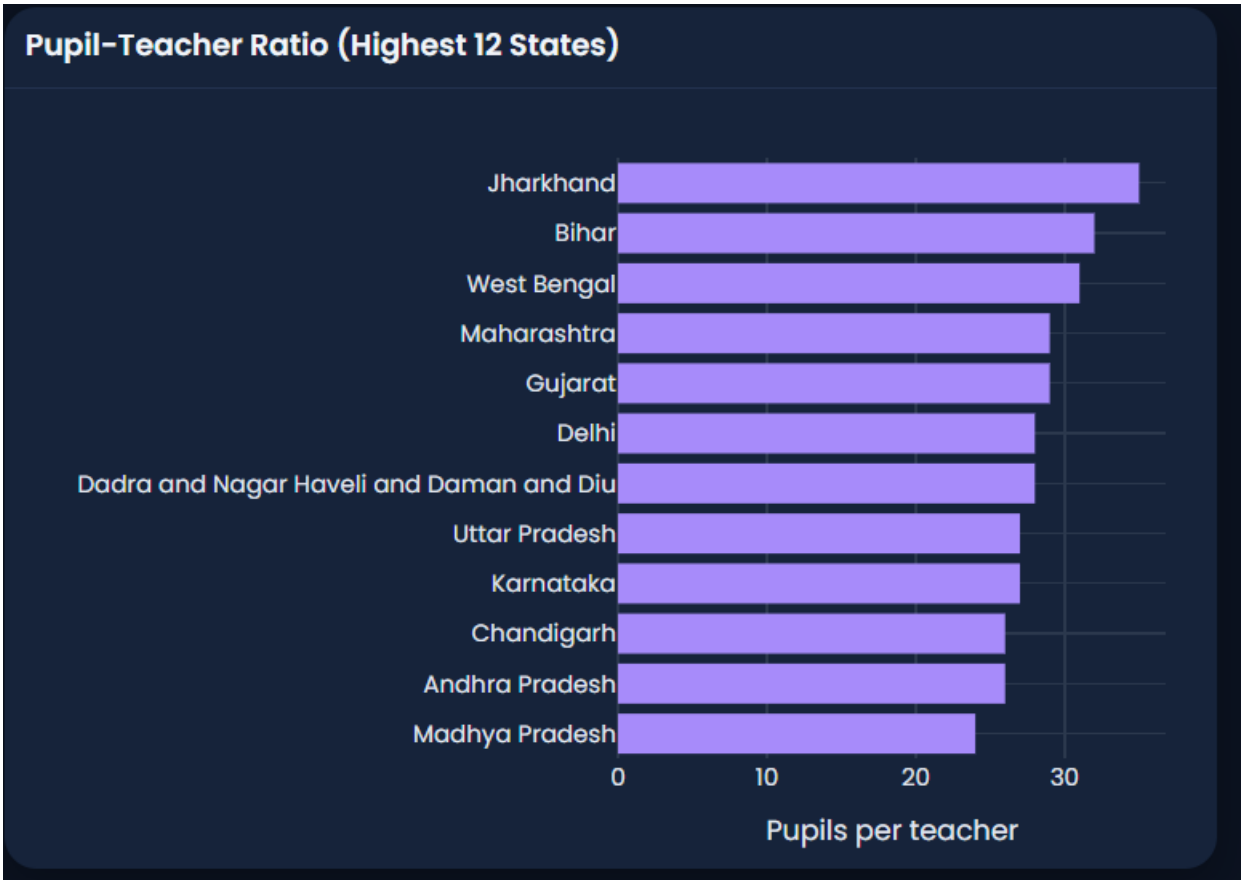


Figure 10: P



upil–teacher ratio for the highest State/UT values.

4.9 Analysis 9: Gender Equality Across Literacy and Enrolment

i. Introduction

Gender equality is examined using both literacy and enrolment measures. These indicators describe different stages of the education pathway and come from different reference years, so their comparison is exploratory rather than causal.

ii. General Description

The project compares female literacy from Census 2011 with girls' share of total enrolment from UDISE+ 2021–22. It also ranks State/UTs by girls' enrolment share and compares the value with a 50% parity reference line.

iii. Specific Requirements, Functions and Formulas

```
Girls_Share_pct = Girls / (Boys + Girls) * 100

ggplot(enrolment_data,
       aes(x = reorder(State, Girls_Share_pct),
           y = Girls_Share_pct)) +
  geom_col() + coord_flip() +
  geom_hline(yintercept = 50, linetype = "dashed")
```

iv. Analysis Results

The calculated overall girls' share across the state-level 2021–22 totals is 48.0%. The highest State/UT share in the supplied table is Meghalaya (51.1%), while the lowest is Haryana (45.2%). A 50% line is a simple composition benchmark; it is not a complete measure of gender equality.

v. Visualization

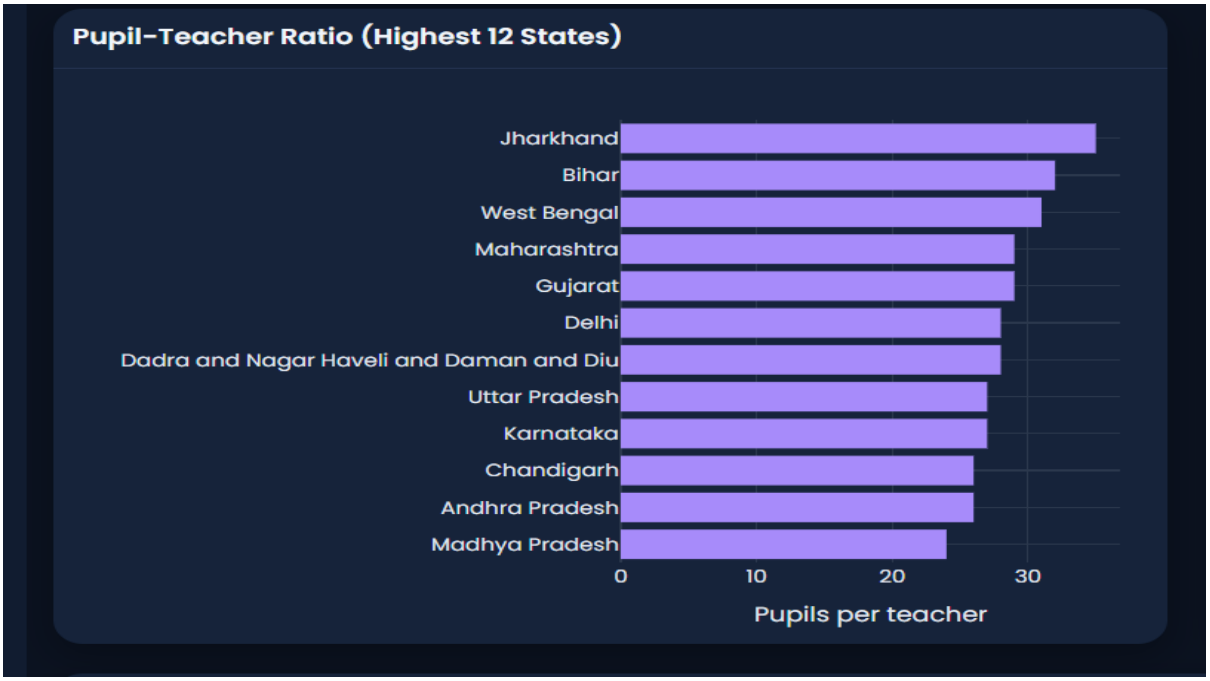


Figure 13: Female literacy (2011) versus girls' enrolment share (2021–22).

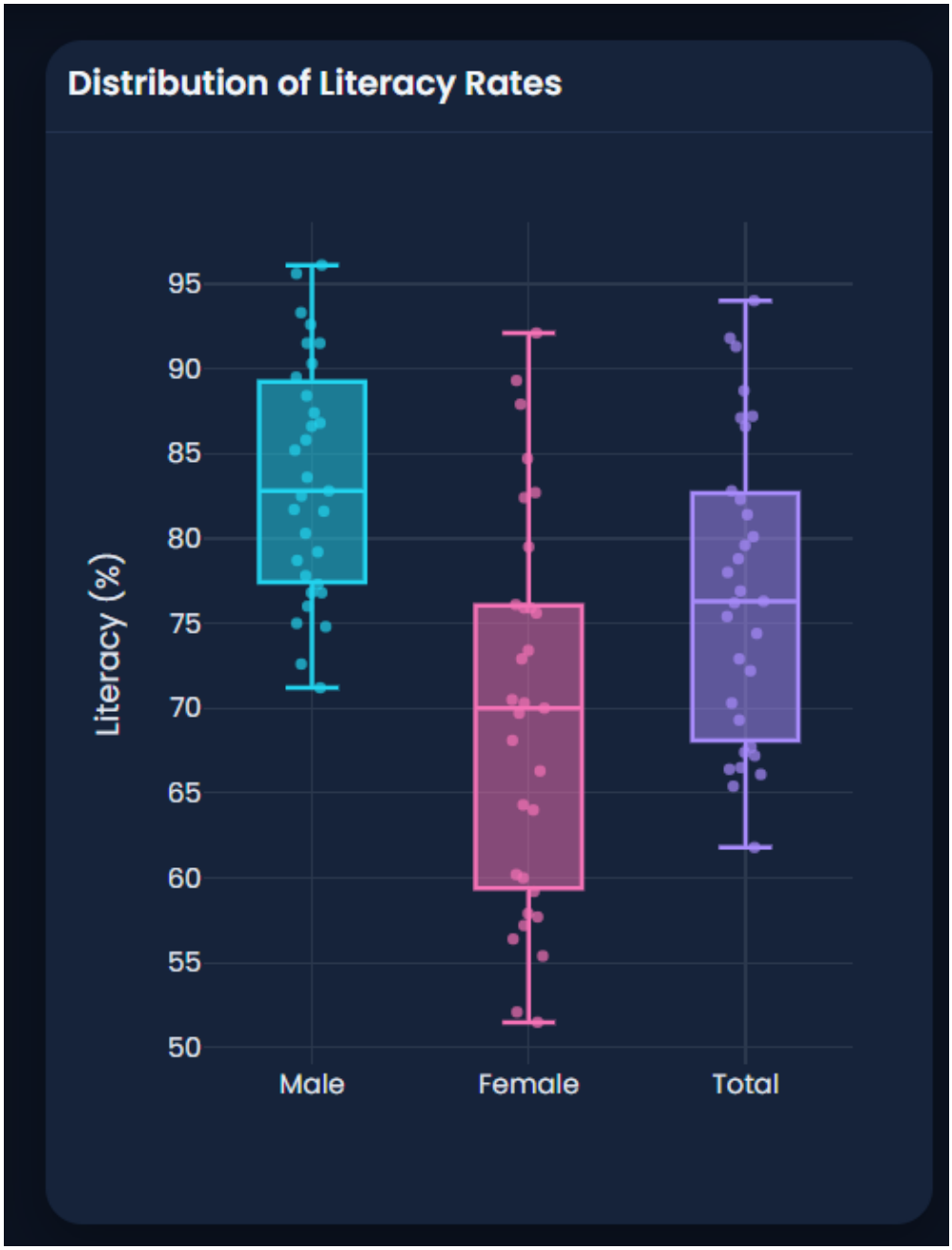


Figure 14: Girls' share of total enrolment by State/UT.

4.10 Analysis 10: Advanced Visualisation — Gender by Stage Heat Map

i. Introduction

A stage-wise gender heat map allows comparison of girls' enrolment share across multiple school stages and States/UTs. It can reveal whether the overall gender composition is consistent across stages.

ii. General Description

The chart uses State/UT on one axis, school stage on the other and girls' share as the colour variable. Hover interaction in the dashboard displays the State/UT, stage and percentage for an individual tile.

iii. Specific Requirements, Functions and Formulas

```
girls_stage_share <- stage_data %>%
  mutate(Girls_Share = Girls / Total * 100)

ggplot(girls_stage_share,
       aes(x = Stage, y = State, fill = Girls_Share)) +
  geom_tile() +
  scale_fill_viridis_c(name = "Girls %")
```

iv. Analysis Results

The heat map shows that girls' share differs by State/UT and school stage. Some State/UT-stage combinations are closer to or above the 50% benchmark, while others are below it. Such differences should be interpreted alongside enrolment totals and the possibility of small denominators.

v. Visualization

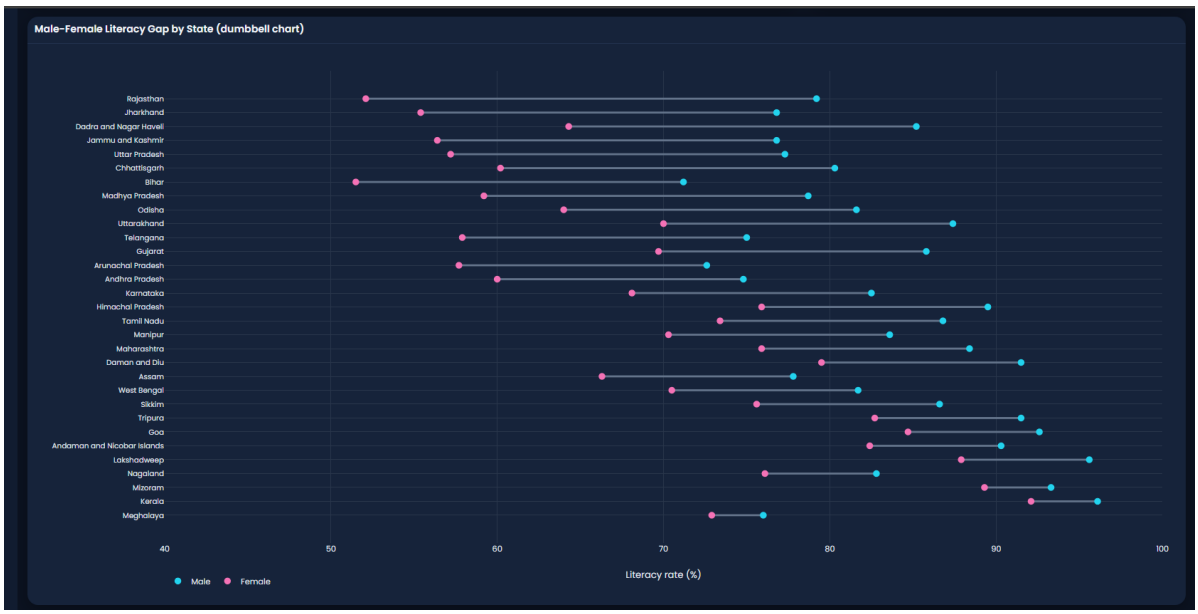


Figure 15: Girls' share by school stage and State/UT (heat map).

4.11 Analysis 11: Word Cloud and Bollinger Band Demonstrations

i. Introduction

The course practical list includes advanced visualisations such as word clouds and Bollinger Bands. These techniques are included as demonstrations, but they do not naturally arise from the available State/UT education tables.

ii. General Description

The word cloud presents education-related terms such as Education, Literacy, Students, Teachers, Schools, Equality and Enrolment. Its word sizes are illustrative and must not be interpreted as measured frequencies from the Census or UDISE+ files. The Bollinger Band chart uses a simulated numeric series to demonstrate a rolling average and volatility envelope.

iii. Specific Requirements, Functions and Formulas

```
# Rolling mean and standard deviation:
Middle_t = mean(x[t-9 : t])
Upper_t  = Middle_t + 2 * sd(x[t-9 : t])
Lower_t  = Middle_t - 2 * sd(x[t-9 : t])

# The series x is simulated for demonstration only.
```

iv. Analysis Results

The word cloud is a communication graphic, not an empirical result. The Bollinger Band is a statistical visualisation demonstration and does not indicate educational volatility. These two outputs are retained only to show the breadth of visualisation methods covered by the practical syllabus.

v. Visualization

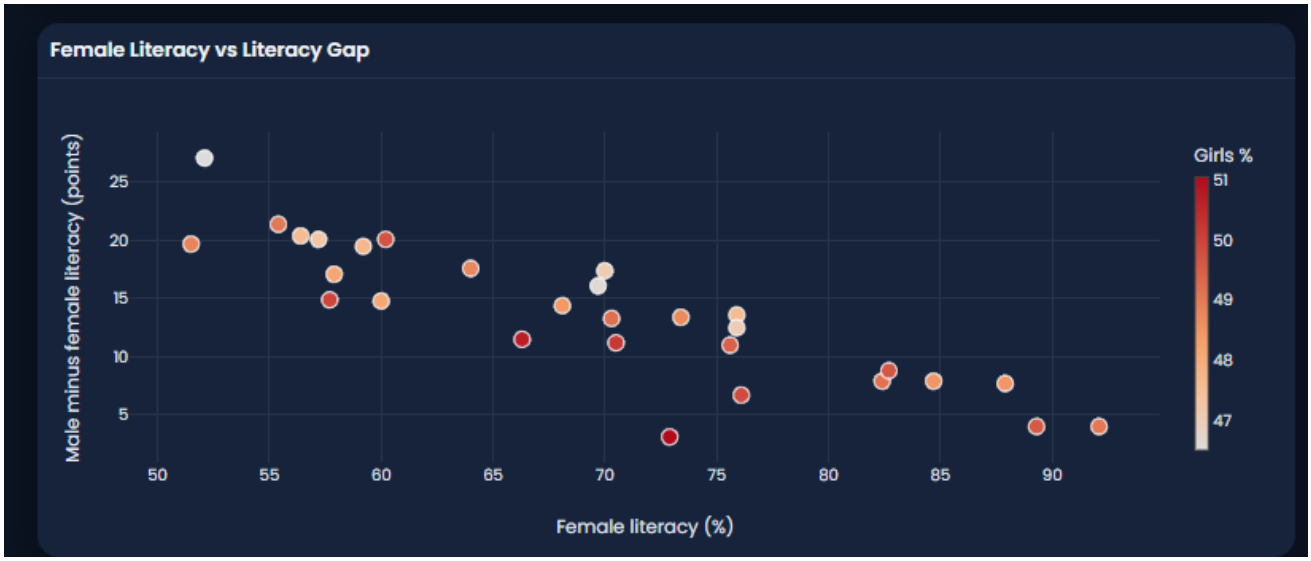


Figure 3: Education-themed word cloud; term frequencies are illustrative.



Figure 12: Bollinger Band demonstration using simulated values.

4.12 Analysis 12: Interactive Visualisations and Dashboard

i. Introduction

Static charts are useful in printed reports, while interactive charts help users explore values and individual State/UT observations. Plotly provides hover labels, zooming and pan controls, and Shiny connects user inputs to reactive outputs.

ii. General Description

The standalone Shiny application contains navigation for Overview, Literacy Analysis, Enrolment Analysis, School Infrastructure, Gender Equality and Data Explorer. It displays summary value boxes, interactive charts and data tables. Plotly outputs support hover-based inspection.

iii. Specific Requirements, Functions and Formulas

```
output$literacy_chart <- renderPlotly({
  plot_ly(
    literacy_data,
    x = ~Male, y = ~Female,
    type = "scatter", mode = "markers",
    text = ~State,
    hovertemplate = paste(
      "State: %{text}<br>",
      "Male: %{x:.1f}%<br>",
      "Female: %{y:.1f}%<extra></extra>"
    )
  )
})
```

iv. Analysis Results

The screenshots supplied with the project show interactive charts for literacy, enrolment by stage, school infrastructure, PTR and gender indicators. The application presents the same core data domains as the report. The screenshots are evidence of the interface appearance; the final submission should include a successful run of the separate R Markdown Flexdashboard if that is explicitly required by the course.

v. Visualization

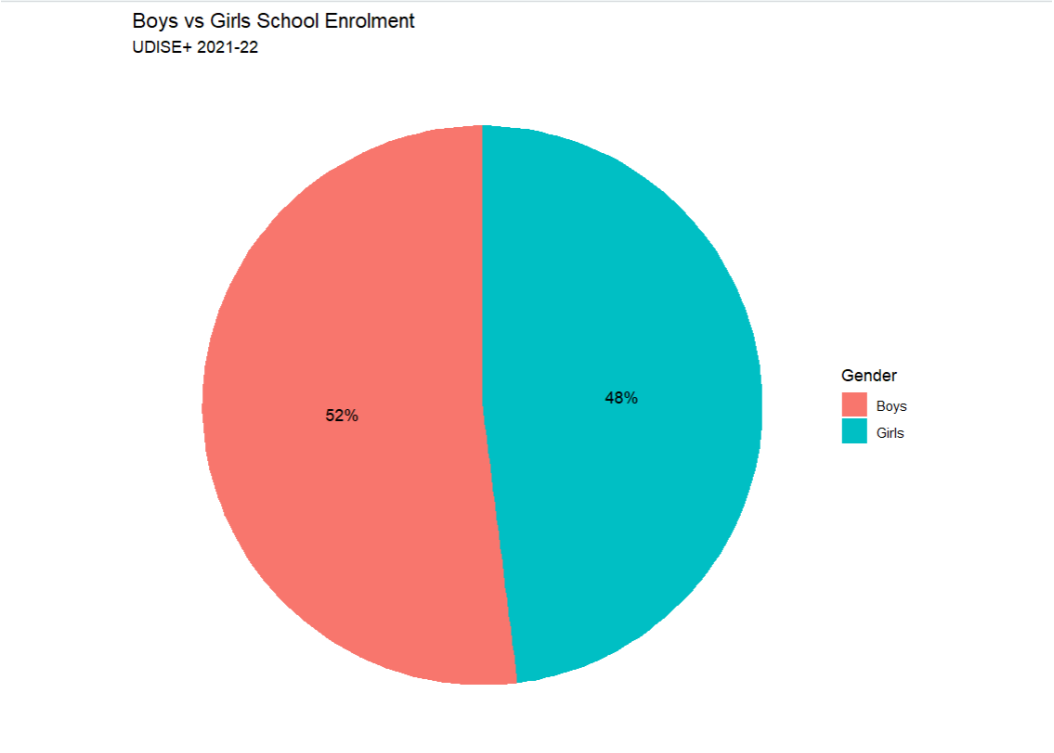


Figure 16: Education Equality in India dashboard interface (captured project view).

5. CONCLUSION

5.1 Summary of Work

The Education Equality in India project demonstrates an end-to-end workflow in R for transforming public education statistics into structured analysis and visual communication. It covers source inspection, preprocessing, data manipulation, descriptive summaries, visualisation and dashboard presentation.

The literacy analysis highlights differences between male and female literacy across States/UTs. The enrolment analysis shows the gender composition of school participation and its variation across school stages. The infrastructure analysis summarises school, enrolment and teacher counts and examines pupil–teacher ratios, zero-enrolment schools and single-teacher schools.

5.2 Verified Infrastructure Summary

Indicator	Value from supplied raw table
State/UT records after excluding India aggregate	36
Total schools	1,471,891
Total enrolments	248,045,828
Total teachers	9,807,600
Unweighted mean State/UT PTR	21.2

These totals are calculated by summing the State/UT rows after excluding the national aggregate. They should be checked against the final processed CSV and any official source revision before submission.

5.3 Key Findings

- The literacy file contains 31 State/UT records after removal of the India aggregate. Female literacy varies substantially across the observations.
- The highest female literacy observation in the supplied Census table is Kerala (92.1%), while the lowest is Bihar (51.5%).
- The mean State/UT male-minus-female literacy gap is 13.9 percentage points.
- The UDISE+ 2021–22 aggregate enrolment split in the dashboard is approximately 52% boys and 48% girls.
- The stage-wise chart shows Primary enrolment as the largest displayed stage, with lower counts at later stages. These counts are not cohort transition rates.
- The 2023–24 infrastructure file totals 1,471,891 schools, 248,045,828 enrolments and 9,807,600 teachers across 36 State/UT rows.
- The highest PTR observation is Jharkhand (35.0); State/UT PTR values should be interpreted with their underlying pupil and teacher counts.

Overall, the results show why education equality should be assessed through multiple indicators. A near-balanced national enrolment composition does not imply that all States/UTs or school stages have achieved parity, and a high number of schools does not automatically imply adequate staffing or equal access.

6. FUTURE SCOPE AND LIMITATIONS

6.1 Limitations

Limitation	Implication for interpretation
Different reference years	Literacy (2011), enrolment (2020–21/2021–22) and infrastructure (2023–24) are not a same-year panel.
State/UT aggregate data	Within-State inequalities and individual student experiences are not visible.
Counts depend on population size	Large States can have larger counts without having better access per child.
PTR aggregation	Unweighted mean State/UT PTR is not a national weighted ratio.
No causal design	Associations do not establish that one indicator causes another.
Demonstration charts	Word cloud and Bollinger Band are illustrative/simulated, not education outcome evidence.
Administrative changes	State/UT names and boundaries may differ across source years.

6.2 Future Scope

- Add district-level data to identify within-State disparities.
- Use population denominators and age-specific enrolment rates rather than counts alone.
- Add attendance, transition, retention, dropout and learning-outcome indicators.
- Create a comparable multi-year panel using consistent definitions and territorial boundaries.
- Calculate weighted national PTR from pupil and teacher totals where source definitions permit.
- Add statistical tests only after stating hypotheses, assumptions and sample sizes.
- Develop a reproducible R Markdown Flexdashboard with reactive filters and downloadable tables.
- Add accessibility features, explanatory tooltips and source-year labels to every dashboard page.

7. REFERENCES

1. Office of the Registrar General & Census Commissioner, India. Census of India 2011. Literacy tables by State/Union Territory and sex. Government of India.
2. Department of School Education and Literacy, Ministry of Education, Government of India. UDISE+ Report 2020–21. Enrolment tables.
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5. R Core Team. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.
6. Wickham, H. et al. tidyverse: Easily Install and Load the 'Tidyverse'. R package documentation.
7. Wickham, H. ggplot2: Elegant Graphics for Data Analysis. Springer.
8. Sievert, C. Interactive Web-Based Data Visualization with R, plotly, and shiny. Chapman and Hall/CRC.
9. Chang, W. et al. shiny: Web Application Framework for R. R package documentation.
10. RStudio / Posit. RStudio IDE and Shiny documentation.
11. DataTables. DataTables JavaScript library documentation; accessed through the R DT package.

Note: Before final submission, add the exact official report URLs, access dates and any specific table numbers from the downloaded source publications. The raw CSV filenames supplied with the project are listed in Chapter 2.

APPENDIX A: GitHub & Linkdein

APPENDIX B: REPRODUCIBILITY AND REQUIREMENT MAPPING

A.1 Project Requirement Coverage

Required component	Report location / evidence
Data Cleaning and Preprocessing	Chapter 3.3–3.4; text trimming, numeric conversion, missingness and duplicate checks
Data Manipulation using R	Chapter 3.5 and analysis formulas using filter, mutate, arrange, summarise and pivot_longer
Exploratory Data Analysis (EDA)	Chapter 3; summary statistics, distribution checks and state rankings
Statistical Analysis	Descriptive means, sums, gaps, shares and PTR analysis in Chapters 3–5
Basic and Advanced Visualizations	Pie, bar, line, scatter, heat map, word cloud, donut and Bollinger Band demonstration
Interactive Visualizations using ggplot2/Plotly	Chapter 4.12; ggplot2 and Plotly hover/zoom
Interactive Dashboard using R Markdown	Prepare and submit education_dashboard.Rmd / rendered HTML separately; app.R is standalone Shiny
Key Findings and Conclusion	Chapter 5

A.2 Reproducibility Checklist

- Keep raw CSV files unchanged in data/raw/ and cleaned files in data/processed/.
- Record package versions and the date the source data were downloaded.
- Run preprocessing scripts before analysis and dashboard scripts.
- Confirm the India aggregate row is excluded from State/UT-only sums.
- Recalculate summary values from the final cleaned data before submission.
- Capture screenshots from the final working dashboard and use the same chart captions as this report.
- Render the R Markdown Flexdashboard successfully and include its HTML/PDF output if requested.

Student details, supervisor details, semester information and final page numbering must be completed before printing.